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SCHOOL OF ELECTRICAL ENGINEERING AND COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAM OF SOFTWARE ENGINEERING

Offline Amharic First Aid Advisor Using NLP and Offline Maps

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DECLARATION

We are students of Adama Science and Technology University in the School of Electrical Engineering and Computing in the Department of Computer Science and Engineering. The information found in this project is our original work. And all sources of materials that will be used for the project work will be fully acknowledged.

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This project has been submitted for examination with our approval as a university advisor.

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Acronym

NLP	Natural Language Processing
GPS	Global Positioning System
API	Application Programming Interface
UI	User Interface
UX	User Experience
AI	Artificial Intelligence
DB	Database
ML	Machine Learning
UAT	User Acceptance Testing
SRS	Software Requirement Specification
JSON	JavaScript Object Notation
STT	Speech To Text
OS	Operating System
IDE	Integrated Development Environment

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Abstract

In Ethiopia, it might be hard to get reliable first aid information, especially in places where the internet is slow and people don't speak the same language. During emergencies, people struggle to find accurate, understandable, and immediate guidance before professional help arrives. Existing first aid applications rely heavily on internet access and English content, making them less effective for local users.

This project aims to develop an Offline Amharic First Aid Advisor App that uses Natural Language Processing (NLP) and offline mapping technologies to provide quick, accessible, and localized emergency assistance. The main objectives are to: deliver step-by-step first aid guidance in Amharic through text and audio, integrate an offline map showing nearby hospitals and first aid organizations, and include visual (photo/video) guides for better understanding.

The system's scope focuses on common emergencies such as bleeding, burns, fractures, and choking. It employs mobile development tools, an offline NLP model, and a local database for map and content storage.

This project is significant because it enhances public health awareness, reduces preventable deaths, and empowers users to act effectively during emergencies without depending on internet access making it a vital innovation for Ethiopia's healthcare accessibility.

CHAPTER 1: INTRODUCTION

1.1 Introduction

In many developing countries, especially across Sub-Saharan Africa, access to timely and accurate first aid information remains a critical challenge. Ethiopia, with its diverse population and vast rural settlements, is no exception. People in both urban and rural communities frequently face medical emergencies such as bleeding, burns, fractures, choking and other injuries, yet access to professional medical assistance is often delayed due to distance, lack of transportation, or limited communication services. In such scenarios, the ability of individuals to provide correct first aid can determine the difference between life and death. However, the majority of available first aid resources are text-based, English-only and dependent on internet connectivity making them inaccessible to the broader Ethiopian population, particularly those who primarily speak Amharic and live in offline environments

This project, Offline Amharic First Aid Advisor Using Natural Language Processing and Offline Maps, is developed to address this accessibility gap by providing an intelligent, offline-capable mobile application that guides Ethiopian users in emergency situations. The application is designed to work offline, ensuring functionality in areas with poor or no internet connectivity. Through the integration of Amharic Natural Language Processing (NLP), the system allows users to ask questions or describe emergency situations using natural, everyday Amharic expressions. The model then interprets the input and provides accurate, step-by-step first aid instructions tailored to the emergency.

Furthermore, the system includes offline map functionalities that allow users to locate nearby hospitals, clinics or health centres without needing internet access. This ensures that after administering initial first aid, users can quickly find the nearest medical facility for professional treatment.

The overall purpose of this project is to create a reliable, accessible and user-friendly first aid support system that empowers Ethiopian communities, reduces preventable injuries and enhances emergency response. The project combines advanced technologies such as TensorFlow Lite for NLP, Flutter for cross-platform development, SQLite for offline data storage, and Mapbox/Leaflet for offline maps with locally relevant content designed in Amharic, Ethiopia's most widely spoken language.

1.2 Background of the Project

First aid is the immediate assistance provided to an injured or ill person before professional medical help becomes available. It plays a crucial role in stabilizing a patient, preventing the condition from worsening, and sometimes even saving lives. However, in many low resource settings, first aid knowledge is limited, and access to information is highly restricted due to language barriers, low literacy levels and lack of digital infrastructure.

In Ethiopia, most of the population resides in rural or semi-urban areas where emergency medical services are limited and internet connectivity is unreliable. Most available first aid mobile applications require an active internet connection, are written in English, and are designed for high income regions. This makes them impractical for the average Ethiopian user.

The absence of Amharic-language first aid resources presents a significant challenge. While some hospitals and health organizations provide printed manuals, these are often text-heavy, outdated, or unavailable during emergencies. Moreover, many people struggle to search for specific first aid procedures on the internet due to language and literacy limitations.

Recent advancements in mobile technology and AI specifically NLP and offline storage techniques have opened new opportunities for health education and emergency assistance. NLP models can process human language, understand user intent, and provide relevant responses even in low-resource languages like Amharic. With the emergence of lightweight models such as TensorFlow Lite, running NLP directly on mobile devices without internet access has become feasible.

This project builds upon these technological advancements to address Ethiopia's unique challenges. The tool combines:

- Offline Amharic NLP for emergency query understanding

- Offline databases for first aid instructions and health centre information
- Offline maps to show nearby medical facilities
- Audio, text, and image guidance for accessibility

This project is motivated by the need to reduce preventable deaths caused by slow or incorrect first aid response. While existing systems offer basic first aid information, they have limitations:

- Limitations of Existing Systems
- Require internet connection
- Are written in English

Provide general information rather than guided step-by-step emergency instructions

Lack interactive features such as NLP based question answering

Do not provide offline hospital location services

Thus, there is a gap in the Ethiopian context for a solution that is offline, Amharic-based, intelligent and user-friendly. This project aims to fill that gap.

1.3 Statement of the Problem

Medical emergencies can occur anywhere and at any time. In Ethiopia, many injuries including bleeding, burns, fractures, choking or poisoning are often mishandled due to lack of immediate knowledge or assistance. According to community health assessments, thousands of preventable injuries lead to complications or fatalities because the affected person does not receive proper first aid before reaching a healthcare facility.

The major problem arises from:

- Language Barrier – Existing first aid apps are mostly in English, making them difficult for Amharic-speaking users.
- Lack of Internet Connectivity – Many regions lack stable internet access, making online applications impractical during emergencies.
- Limited Awareness – People lack basic first aid training and depend on hospitals that may be far away.
- Lack of Localized, Step-by-Step Guidance Many available resources are general, textual, or non-interactive.

- Difficulty Locating Nearby Medical Facilities-Without the internet, users cannot search for nearby hospitals or clinics.

As a result, communities are unable to respond effectively during medical emergencies. Delays in providing first aid can worsen the injury or even lead to death.

This project addresses the following core problem:

There is no intelligent, offline, Amharic-language first aid mobile application in Ethiopia that can guide users step-by-step during emergencies and help them locate nearby medical facilities without internet access.

1.4 Objective of the Project

1.4.1 General Objective

The general objective of this project is to design and develop an offline, Amharic-language first aid advisor mobile application that uses Natural Language Processing and offline maps to provide emergency guidance and help users locate nearby medical facilities without internet connectivity.

1.4.2 Specific Objectives

The specific objectives break down the project into measurable and achievable tasks:

- To design an Amharic NLP model capable of understanding user queries related to first aid emergencies.
- To collect, design, and organize first aid instructions for common emergency scenarios, including bleeding, burns, fractures, choking, and other injuries.
- To implement an offline database (SQLite) to store instructions, multimedia content, and hospital information.
- To develop a mobile application using Flutter that supports offline operations and user-friendly interfaces.
- To integrate offline maps to display nearby hospitals, clinics or health centres.
- To create audio, text and visual first aid guidance for users with low literacy levels.
- To test the system for performance, accuracy, usability and reliability.

- To prepare complete technical and user documentation for the final project submission.
- To evaluate the effectiveness of the app in providing timely first aid guidance and improving user emergency response.

1.5 Scope and Limitations

This section defines the boundaries, assumptions, and constraints of the Offline Amharic First Aid Advisor Using NLP and Offline Maps project. It explains what the system is designed to accomplish, the target users and their needs, and the factors that may influence the performance, usability, and reliability of the application. The scope highlights the functionalities, technologies, and emergency scenarios the app will cover, while the limitations acknowledge technical, operational, and resource-related challenges that may affect the project's implementation and outcomes.

1.5.1 Scope of the Project

The Offline Amharic First Aid Advisor is designed to help people in Ethiopia access immediate first aid guidance, even without internet access. The main points of the project include:

1. Target Users

- **Everyday Users:** Amharic-speaking individuals who may face emergencies and need quick, reliable guidance.
- **Healthcare Workers and Educators:** People who can use the app as a teaching or awareness tool.
- **Developers and Researchers:** Those interested in offline AI, natural language processing for Amharic, or mobile healthcare solutions.

2. Functional Features:

- Provide step-by-step first aid instructions for common emergencies, including bleeding, burns, fractures, choking, and minor injuries.
- Multiple formats: text, audio, and images/videos to make it accessible for everyone, including people with different literacy levels.

- Amharic NLP (Natural Language Processing) to understand typed or spoken questions and provide accurate first aid advice.
- Integrate offline maps showing nearby hospitals, clinics, and first aid centers, allowing users to locate medical assistance without internet connectivity.
- Extra features such as emergency call buttons and interactive tutorials.

3. Non-Functional Features

- Ensure reliability and performance in offline mode, so it works even in areas with no internet.
- Lightweight and optimized for mobile devices with limited storage.
- Design an intuitive and user-friendly interface suitable for a wide range of users, including non-technical and rural populations.
- User privacy and security are carefully considered, especially for location-based features.

4. System Architecture Development Methodology:

- The system will follow a mobile-first, offline-first architecture, integrating modular components for NLP, mapping, content delivery, and user interface.
- Development will employ Agile methodology, allowing iterative testing, feedback incorporation, and progressive enhancement of offline features and user interaction.

5. Evaluation Criteria

- Effectiveness of emergency guidance in simulated scenarios.
- Accuracy of NLP-based emergency classification.
- Usability and accessibility for Amharic-speaking users.
- Performance and responsiveness in offline environments.

1.5.2 Limitations of the Project

Despite its scope and intended functionality, the project has several limitations and potential challenges:

1. Technical Limitations:

- NLP for Amharic, a low-resource language, may face challenges in accurately interpreting diverse user inputs, slang, or non-standard expressions.
- Offline maps and multimedia content may increase storage requirements, limiting usability on low-end devices.
- AI features may slow down some devices or drain the battery faster.

2. Operational Limitations:

- The app provides first aid guidance only and cannot replace professional medical care. Users may still face risks if the instructions are followed incorrectly.
- Effectiveness may be limited in rare or complex emergencies that fall outside the predefined scenarios.

3. Economic and Resource Constraints:

- Development relies on local resources, datasets, and libraries, which may be limited for Amharic NLP and offline mapping functionalities.
- High-quality multimedia content creation may require additional time and effort, impacting project timelines.

4. Ethical Considerations:

- Ensuring accurate, safe, and culturally appropriate guidance is critical, as incorrect advice may cause harm.
- Privacy of location data must be maintained, even in offline mode, to protect user confidentiality.

5. Dependency Risks:

- The system depends on preloaded map data and local databases; any missing or outdated data may affect accuracy.
- Device compatibility and performance may vary depending on hardware limitations.

6. Other Risks:

- Offline databases need regular updates; outdated information can reduce reliability.
- Device compatibility may limit some features on older smartphones.

By clearly defining the scope and limitations, this project establishes realistic expectations, identifies potential risks, and provides a structured framework for development and evaluation,

ensuring that the Offline Amharic First Aid Advisor remains both practical and achievable within the graduation project context.

1.6 Deliverables

By the end of the project, we aim to deliver:

1. **Mobile Application:** A fully functional Android mobile application that provides offline first aid guidance in Amharic. The app includes text, audio, and visual instruction for common emergencies, as well as offline mapping of nearby hospitals and clinics. The application should operate fully offline, correctly process Amharic inputs through the NLP model, display step-by-step guidance, and provide offline navigation to nearby healthcare facilities
2. **Amharic NLP Engine:** A custom Natural Language Processing (NLP) component capable of interpreting Amharic emergency queries, classifying the type of emergency and delivering appropriate first aid guidance.
3. **Offline Maps and Database Integration:** A local map and content database storing hospitals, clinics, and first aid centers, as well as multimedia content (images, videos, and audio guides).
Users should be able to locate nearby medical facilities and access all multimedia first aid content without internet connectivity, and the database must be optimized for storage and performance on mobile devices.
4. **User Interface Experience Design:** An intuitive and user-friendly interface tailored to Amharic-speaking users, including navigation, emergency selection, and content display. The interface must be accessible, culturally appropriate, and easy to navigate, verified through user testing and feedback.
5. **Project Documentation:** Comprehensive documentation including the Software Requirements Specification (SRS), system architecture and design documents, user manuals and guides, and technical implementation details.
Documentation should provide clear instructions and reference material for developers, stakeholders, and end users, ensuring maintainability and usability of the system.

6. **Evaluation and Testing Reports:** Reports detailing the functional and usability testing of the system, including test cases, performance benchmarks, and user feedback analysis. The system must pass all functional tests, meet performance and usability benchmarks, and demonstrate effectiveness in providing offline first aid guidance.
7. **Training Materials(Optional/If Applicable):** Simple guides or video tutorials for users or healthcare educators to understand and use the application effectively. Materials should be clear, concise, and demonstrate how to use all core features of the application.

1.7 Feasibility Study

The feasibility study will evaluate the viability, practicality, and potential benefits of the Offline Amharic First Aid Advisor Using NLP and Offline Maps project. It assesses technical, operational, and economic aspects to determine whether the project can be successfully developed, deployed, and maintained.

1.7.1 Technical Feasibility

The project is technically possible. Smartphones today can handle offline apps with NLP and multimedia content. Open-source tools like Flutter, TensorFlow Lite, and SQLite support development, and the modular design makes integration smooth

Hardware: Modern Android devices are capable of running offline applications with multimedia content, NLP processing, and offline maps. Low-end devices may have limitations, but optimization strategies will be applied to ensure broad compatibility.

Software and Tools: The project will utilize mobile development frameworks (e.g., Flutter or React Native), NLP libraries compatible with Amharic, local databases for offline storage, and mapping APIs for offline routing. All these tools are available and compatible with the system requirements.

System Complexity: While the system integrates multiple components (offline NLP, offline maps, multimedia content, and interactive UI), modular design and Agile development methodology make it technically achievable.

Developer Expertise: The student possesses knowledge in mobile development, NLP, database management, and software design, ensuring that the project is within the technical capabilities of the development team.

Overall, the project is technically feasible as all necessary resources, tools, and expertise are available, and the system architecture is compatible with current mobile technologies.

1.7.2 Operational Feasibility

The project is operationally feasible because it meets the needs of the target users and is designed for practical usability:

- **Usability and Functionality:** The application provides clear, step-by-step first aid guidance in Amharic and offline maps, making it practical and easy to use for individuals in emergency situations.
- **User Acceptance:** Amharic-speaking users are expected to adopt the system easily due to its local language support, multimedia instructions, and offline functionality, which address existing gaps in emergency preparedness.
- **Maintenance and Support:** The system will include modular code and documentation, enabling future updates, content additions, and maintenance without major redesign.
- **Security and Privacy:** The application minimizes risks by storing data locally and using offline maps, ensuring user location and personal data privacy.

Hence, the system is operationally feasible as it aligns with the real-world needs of users and can be maintained and supported effectively.

1.7.3 Economic Feasibility

The project is economically feasible based on the cost-benefit analysis and resource allocation:

- **Development Costs:** The project primarily relies on free and open-source tools such as Flutter, TensorFlow Lite, and open-access datasets for Amharic language processing. Most development, testing, and design activities can be completed using freely available software and personal devices. The main cost is associated with publishing the application on the Google Play Store, which requires a one-time developer account registration fee.
- **Benefits:** The system provides significant social and health benefits, including increased public awareness of first aid, faster response in emergencies, and potential reduction in preventable injuries and fatalities.
- **Resource Utilization:** Existing hardware (laptops, mobile devices) and open-source software libraries reduce overall expenses.
- **Return on Investment:** While the project is primarily academic, the system's deployment could benefit communities, NGOs, and health organizations, creating long-term value.

Overall, the project is economically feasible as the benefits outweigh the costs, and resource allocation is reasonable and sustainable within the constraints of a graduation-level project.

1.8 Significance of the Project

The Offline Amharic First Aid Advisor project will be significant because it will address critical gaps in emergency preparedness, healthcare accessibility, and public safety in Ethiopia. Its value can be examined across multiple dimensions:

- **Contribution to the Problem Domain:** The project directly targets the lack of reliable first aid information in areas with poor internet connectivity and limited healthcare access. By providing offline, Amharic-language guidance, the system enables individuals to respond effectively to emergencies, potentially reducing preventable injuries and fatalities.

- **Benefit to Stakeholders:** The primary stakeholders include patients, community members, healthcare educators, and emergency responders. Patients and community members gain immediate access to first aid instructions in their native language. Healthcare educators and trainers can use the system as a teaching tool, enhancing community health literacy and preparedness.
- **Societal Impact:** The project has a broader social impact by empowering communities to act in medical emergencies, promoting awareness of proper first aid procedures, and enhancing healthcare equity in underserved regions. Its offline-first design ensures that even rural and low-connectivity areas can benefit from life-saving guidance.
- **Academic and Technological Contribution:** From an academic perspective, the project demonstrates the application of Natural Language Processing (NLP) for a low-resource language, offline mobile application design, and system integration of multimedia and mapping functionalities. It contributes original insights into technology-driven healthcare solutions for low-resource settings and serves as a reference for future research and development in similar contexts.

Innovation and Originality: The system's combination of offline functionality, Amharic language support, NLP-based emergency classification, and multimedia guidance represents an innovative approach not currently available in existing first aid applications. It highlights the potential of technology to solve locally relevant healthcare challenges.

1.9. Beneficiaries of the Project

The Offline Amharic First Aid Advisor project has a wide range of potential beneficiaries, all of whom gain direct or indirect advantages from its features and functionality. The main beneficiaries include:

- **General Public / Amharic-speaking Users:**
Individuals in Ethiopia, especially those in rural or low-connectivity areas, benefit directly by gaining instant access to reliable first aid guidance in their native language.

The system empowers them to respond effectively during medical emergencies, potentially reducing injuries and saving lives.

- **Patients and Emergency Victims:**

Individuals who experience accidents, burns, fractures, choking, or other emergencies benefit from step-by-step instructions, audio guidance, and visual aids. This allows immediate self-help or assistance to others before professional medical care is available.

- **Healthcare Educators and Community Health Workers:**

Teachers, trainers, and community health personnel can use the application as a training tool to educate communities on first aid procedures and emergency preparedness. The offline functionality ensures usability in training sessions in areas without internet access.

- **Hospitals and Emergency Response Units:**

While the system does not replace professional care, hospitals and local clinics indirectly benefit as patients receive preliminary first aid before arrival, improving the efficiency of emergency care and reducing complications.

- **Researchers and Developers:**

Academics and software developers interested in NLP for low-resource languages, offline mobile applications, or healthcare technology can use the project as a reference model for further research or system improvements.

- **Society at Large:**

By improving emergency response and first aid awareness across communities, the project contributes to public health improvements and reduces preventable fatalities, creating long-term social benefits.

In summary, the project provides practical, educational, and societal value by delivering a user-friendly, offline, and localized first aid solution that benefits individuals, communities, healthcare professionals, and the broader field of healthcare technology research.

1.10 Methodology

The Offline Amharic First Aid Advisor project will follow a methodology that will combine structured phases with Iterative agile development principles, specifically scrum framework.

This approach ensures systematic progression through major project milestones while maintaining flexibility for adaptive development based on continuous testing and user feedback. The methodology is organized into six sequential phases with an iterative development cycle embedded with the implementation phase.

1.10.1. Planning Phase

The project will begin comprehensive planning activities to establish a solid foundation those include

- Project Initiation to Defining project vision, objectives, and success criteria,
- Stakeholder Analysis to Identify primary users, beneficiaries, and project advisors,
- Feasibility Study to Conduct technical, operational, and economic feasibility assessments,
- Risk Assessment to Identify potential risks and mitigation strategies,
- Resource Planning to Determine required tools, technologies, and team capabilities and
- Timeline Establishment to Create initial project schedule with key milestones

1.10.2. Analysis Phase

This phase will focus on understanding requirements and defining system specifications like

- Requirements Gathering: Conducting research on user needs through literature review and consultations,
- Functional Requirements Analysis for Documenting all system functionalities and user interactions,
- Non-Functional Requirements Analysis to Define performance, reliability, security, and usability requirements,
- System Constraints Identification for Documenting technical, operational, and environmental limitations, and
- Software Requirements Specification (SRS) to Create comprehensive documentation of all requirements

1.10.3. Design Phase

The design phase will translate requirements into technical specifications those are the following

- System Architecture Design: Designing high level system architecture with component interactions,
- Database Design to Create entity-relationship diagrams and define database schema,
- User Interface Design for Developing wireframes, mockups, and prototypes using design tools,
- Technical Specifications for Detail algorithms, data structures, and processing logic, and
- Technology Stack Selection to Finalize all development tools and frameworks

1.10.4. Implementation Phase (Using Agile Scrum)

The implementation will follow an iterative Agile approach with Scrum practices:

- Sprint Planning to Divide work into 2-week sprints with specific deliverable goals
- Backlog Management for Maintaining and prioritizing product backlog items
- Iterative Development:
 - Sprint 0: Setting up environment, foundational code, and initial configurations
 - Feature Sprints: Develop core functionalities incrementally (NLP module, offline maps, content management)
 - Integration Sprints: Combine developed modules into cohesive system
- Daily Stand-ups to Conduct daily 15-minute meetings for progress tracking and issue resolution
- Continuous Integration for Regularly merge code changes with automated testing
- Version Control Use Git for collaborative development and change management

1.10.5. Testing and Quality Assurance Phase

Comprehensive testing will ensure system reliability and effectiveness

- Test Planning to Develop test strategy, test cases, and acceptance criteria
- Unit Testing: Testing individual components (NLP model, database functions, UI elements)
- Integration Testing to Test interactions between integrated modules
- System Testing for Evaluating complete system functionality and performance
- User Acceptance Testing (UAT) to Conduct testing with target users to validate usability and effectiveness
- Performance Testing to Assess system performance under various conditions and device capabilities
- Security Testing to Verify data privacy and system security measures

1.10.6. Evaluation and Deployment Phase

Final assessment and system release will include:

- System Evaluation: Measuring system against defined objectives and success criteria
- Performance Metrics Analysis: Evaluate NLP accuracy, response time, and user satisfaction
- Documentation Finalization: Complete user manuals, technical documentation, and project reports
- Deployment Preparation: Package application for distribution and prepare deployment environment
- Pilot Deployment: Release initial version to limited user group for real-world testing
- Final Presentation: Demonstrate system functionality to stakeholders and academic panel
- Project Closure: Conduct final review, document lessons learned, and archive project materials

1.11. Development Tools

The project will utilize a modern, open-source technology stack optimized for cross-platform development, offline functionality, and embedded AI.

Mobile Application Framework: Flutter (Dart) Chosen for its high performance, single codebase for Android, and rich UI capabilities suitable for offline apps.

Offline AI & NLP Engine:

- TensorFlow Lite or ONNX Runtime: For deploying a lightweight, pre-trained Amharic NLP model on the device.
- Transformers Library (Hugging Face) or Scikit-learn: For initial model training and development (Python environment).

Offline Mapping:

Mapbox GL Native or Leaflet with Offline Plugins: SDKs that support storing vector/raster tiles locally for full offline map display and navigation.

Local Database: SQLite. An embedded, lightweight database ideal for storing relational data like first aid steps, hospital locations, and multimedia metadata on the device.

Integrated Development Environment (IDE): Android Studio and Visual Studio Code with the Flutter/Dart plugin.

Design & Prototyping: Figma for creating and testing UI/UX wireframes and prototypes.

Version Control & Collaboration: Git with GitHub for source code management and team collaboration.

Testing Tools:

- Flutter DevTools & Emulator: For debugging and UI performance testing.

- Postman: For testing data update endpoints used during development.

Documentation: Google Docs for SRS, and Draw.io / Lucidchart for system architecture and data flow diagrams.

1.12. Required Resources with Cost

This section estimates the resources and costs that the project requires, such as the human, material, financial, or time resources, or the equipment, infrastructure, or overhead costs.

Resource Category	Item / Description	Purpose	Estimated Cost (ETB)	Notes
Hardware	Laptop/PC (5 units)	Primary development machine for each team member.	0	Assumed to be pre-owned by team members.
	Android Smartphones (Multiple models)	For testing the application on various devices and Android versions.	0	Assumed to be pre-owned by team members.
	Internet Connectivity	For research, downloading tools/libraries, and collaboration during	1,000	Estimated data subscription cost for the project duration.

		development sprints.		
Software & Services	Google Play Developer Account	One-time registration fee to publish the final application.	3,500	Primary anticipated software cost.
	Mapbox/Leaflet Services	Potential cost for downloading and packaging high-quality offline map tiles for Ethiopia.	1,000 (contingency)	Free tiers will be used initially; cost may apply for final map data packaging.
Operational & Materials	Printing & Documentation	Binding of final project reports, user manuals, and presentation materials.	1,000	
Miscellaneous/Contingency	—	Unforeseen expenses, transportation, or small supplies.	500	Buffer for unexpected costs.

Table 1 Required Resources with Cost

Total Estimated Budget

= 6,0000ETB

1.13. Task and Schedule

This section defines the tasks or activities that the project performs or completes, and the schedule or timeline that it follows or meets, such as the work breakdown structure, the task dependencies, the milestones, or the deadlines.

The project follows an Agile Scrum methodology organized into two-week sprints within six major phases. The schedule spans from November 2025 to June 2026 and is structured as follows:

Phase	Sprint	Timeline	Key Objectives & Deliverables
Phase 1: Planning	Sprint 0	1 week	Initiating project, stakeholder analysis, feasibility study, risk assessment, resource planning, timeline establishment. Final Product Backlog creation.
Phase 2: Analysis	Sprint 1	3 weeks	Gathering requirements, analyzing functional and non-functional requirements, system constraints identification, SRS documentation completion.
Phase 3: Design	Sprint 2	1 weeks	Designing system architecture, creationing database schema, UI/UX wireframing and prototyping, technology stack finalization.
Phase 4: Implementation	Sprint 3	1 week	First Aid Content Module: Implementing offline database structure and develop UI for text/image guides for 3 to 5 emergency types.

	Sprint 4	1 week	Complete first aid content for all core emergencies. Integrate audio playback functionality and implement manual search feature.
	Sprint 5	3 weeks	Amharic NLP Foundation: Developing and training initial intent classification model. Integrate TensorFlow Lite with basic keyword processing in the app.
	Sprint 6	1 week	Enhance NLP model and build chatbot UI interface. Connect NLP output to display relevant first aid guides.
	Sprint 7	2 weeks	Offline Mapping Foundation: Integrate Mapbox/Leaflet SDK. Display static offline maps and plot hard to coded hospital locations.
	Sprint 8	1 week	Implement dynamic hospital data loading from the database. Add offline search ("hospitals near me") and basic route display.
	Sprint 9	1 week	System Integration: Integrate NLP, Content, and Map modules into a unified system. Produce first internal alpha build.
	Sprint 10	1 week	Conducting internal testing and refinement. Fix bugs and optimize performance based on initial feedback.

Phase 5: Testing	Sprint 11	1 week	User Acceptance Testing (UAT): Preparing test protocols and recruit beta testers. Deploy beta version for structured feedback collection.
	Sprint 12	1 week	Implement critical feedback from UAT. Conduct performance and security testing. Finalize all application content.
Phase 6: Deployment	Sprint 13	1 week	Documentation & Finalization: Complete user manual, technical documentation, and final project report.
	Sprint 14	1 week	Deployment Preparation: Prepare final APK and Google Play Store listing. Conduct final performance optimization.
	Sprint 15	1 week	Release: Submit application to Google Play Store. Prepare final project demonstration and presentation materials.
	Sprint 16	1 week	Project Closure: Final presentation to stakeholders and academic panel. Project retrospective and report submission.

Table 2 Task and Schedule

Key Milestones

- **Milestone 1 (End of Phase 3):** Design Phase Completion ~ All architecture, UI/UX, and technical specifications finalized.

- **Milestone 2 (End of Sprint 6):** Integration functional Amharic NLP Chatbot ~ Users can query emergencies in Amharic and receive responses.
- **Milestone 3 (End of Sprint 8):** Offline Mapping & Navigation Functional ~ Complete offline map display with hospital search and directions.
- **Milestone 4 (End of Sprint 9):** First Integrated Alpha Build ~ All core modules working together in a single application.
- **Milestone 5 (End of Sprint 11):** User Acceptance Testing Complete ~ Feedback collected and prioritized from target users.
- **Milestone 6 (End of Sprint 16):** Application Published and Project Closed~ Final version available on Google Play Store with complete documentation

1.14. Team Composition

This section describes the team or group that works on the project, and the roles or responsibilities that they have, such as the project manager, the system analyst, the system designer, the system developer, the system tester, or the system evaluator.

The Offline Amharic First Aid Advisor project is developed by a team of five Software Engineering students with complementary technical skills and domain knowledge. The team follows an Agile methodology where members collaborate closely and share responsibilities across sprints, while each member maintains primary accountability for specific roles as outlined below.

No.	Name	ID	Primary Role & Responsibilities
1.	Nathanael Wondwosen	UGR/25457 /14	Project Manager & Backend Co-developerCoordinate Agile sprints and facilitate daily stand-upsManage product backlog and sprint planningOversee project timeline and milestone trackingAssist in backend development and offline map

			integration Design and implement the offline AI/NLP engine
2.	Bereket Samuel	UGR/25796 /14	Backend Developer & System Analyst Design and implement the offline AI/NLP engine Develop the local database schema and content management system Analyze system requirements and ensure technical feasibility Conduct backend module testing and optimization
3.	Kenasa Tariku	UGR/25996 /14	Frontend Developer & UI/UX Designer Lead Flutter application interface development Design user experience flows and create UI prototypes Implement responsive and accessible Amharic language interfaces Conduct usability testing and gather user feedback
4.	Misgana Tsegaye	UGR/25432 /14	Frontend Developer & System Tester Develop interactive UI components and multimedia integration Design and execute comprehensive testing protocols Coordinate user acceptance testing with target users Ensure cross-device compatibility and performance optimization
5.	Muaz Jemal Hajje	UGR/22544/13	System Designer & Evaluator Design overall system architecture and integration patterns Evaluate technical decisions and system performance

			metricsEnsure compliance with non-functional requirementsAssess security and privacy implementations
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Table 3 Team Composition

Chapter 2 : Description of existing system

This chapter describes how current first-aid information systems work, who uses them, their drawbacks in the Ethiopia/Amharic context, and the main findings from the academic and technical literature about first-aid apps, offline mapping and NLP for low-resource languages.

2.1. Major function of existing system

Modern first-aid mobile apps give people quick and reliable emergency guidance directly on their phones. Their main purpose is to provide short, easy-to-follow instructions for different first-aid situations. This information is usually organized by common scenarios such as bleeding, burns, choking, or unconsciousness, so users can quickly find what they need during an emergency.

2.1.1 What existing apps do

Based on the research we did, many official organizational apps (for example IFRC's First Aid app, the British Red Cross First Aid app and other First Aid apps) present core guidance as short ordered steps, often augmented by brief explanatory text or reminders of priorities (what to do first).

In addition to the textual checklists, some international apps commonly include multimedia elements—illustrations, photos, animations and short videos—that demonstrate techniques (for example, hand placement for chest compressions or how to apply pressure to a wound). These media assets are integrated with the text guidance so users can switch between reading and visual guidance during an emergency.

Some apps also include interactive learning features such as quizzes, scenario-based training modules, and progress tracking so users can practice skills and assess their knowledge over time. This transforms some first-aid apps into both on-demand guidance tools and lightweight training resources.

2.1.2. How do they work

Most existing first-aid apps use a very straightforward interaction flow so users can quickly find the help they need during an emergency. In most cases, the app presents a list of common injuries and emergency situations such as burns, bleeding, choking, fractures, allergic reactions, or unconsciousness. Users usually:

1. **Browse the list of injuries or categories** shown on the home screen.
2. **Tap on the condition** that matches the emergency they are facing.
3. **View the step-by-step instructions** for that specific injury. These instructions often include short text explanations, diagrams, images, or short videos that show what to do.
4. **Scroll through the guidance** to follow the recommended actions in the correct order.
5. **Optionally use the search bar** (in apps that support it) to quickly look for a specific condition if it is not immediately visible in the list.
6. **Use quick-access buttons** like “Call Emergency Services” if the app provides them.

Some apps also include favorites/bookmarks, allowing users to save the most important or frequently used guides so they can open them faster. Other apps also provide a search function for users who prefer to type in the name of an injury instead of browsing through categories. The search bar helps users quickly find specific conditions if the list is long or if they do not know which category an injury belongs to.

Overall, the flow is simple: **open the app** → **choose the type of injury** → **follow the instructions**. This design helps reduce confusion and ensures users can act quickly and confidently during emergencies.

Many apps come with a pre-loaded data and provide offline access to users:

- Some first-aid apps come with all essential instructional text, images, and media bundled into the app itself (or downloaded to the device ahead of time). This means that users can access full first-aid guidance even when there is **no cellular or Wi-Fi connection** — which is crucial in remote areas or during disasters when networks might be down.
- In these apps, once installed (or once the data is downloaded), all content is available locally. Users can simply open the app, select the injury or emergency type, and read or view the instructions immediately — no waiting, no loading screens, no risk of failing due to connectivity issues.

Other apps work online or mixed with and without internet connection:

- Some first-aid apps rely on network connectivity either for initial setup, content downloads, or for retrieving additional resources (updated guides, multimedia, location data, or hospital/clinic lookups). For these apps: when the app is first installed, or when a user selects a scenario not yet stored on the device, the app must connect to the server to download the necessary content. This implies the user needs a working internet connection at least once.
- If the user is in a location without internet (or if their connection is lost), certain content or features may fail to load — for example, images, videos, or external resources may not be available, or hospital-location features may not work.
- Some apps offer a hybrid approach: core emergency guidance is preloaded or cached, but media assets (videos or detailed illustrations) or extra features (maps, nearby hospital search) require internet. In those cases, the user may still get basic first-aid steps offline, but richer content only when online.

2.2. Users of current system

Existing first-aid mobile applications serve a wide range of users. These users can be grouped into several main categories based on how they interact with the system and what type of support they require.

2.2.1 Primary Users (General Public / Laypersons)

These are ordinary individuals with no medical training who rely on first-aid apps during emergencies. They use the system to quickly access step-by-step instructions for injuries such as bleeding, burns, choking, fractures, or fainting. Their main interaction with the system is selecting an emergency condition and following the provided guidance. They require simple, clear instructions and quick access to the steps they need.

2.2.2 Secondary Users (Health Workers, Teachers, and Caregivers)

These users have some basic knowledge of first aid and use existing apps as a reference tool or educational aid. They may use the app to demonstrate first-aid procedures, explain steps to others, or review guidance when assisting someone in need. They expect reliable and accurate information that can be easily used for teaching or support.

2.2.3 Institutional Users (Organizations and Health Agencies)

Institutions such as NGOs, schools, health organizations, and emergency services use first-aid apps to support public safety programs, training, and awareness campaigns. They rely on the system to provide standardized instructions that follow recognized first-aid guidelines. These users often promote the app to communities as a trusted source of emergency information.

2.2.4 Administrative and Technical Users (App Developers and Content Managers)

These users are responsible for updating and maintaining the system. They ensure that first-aid instructions follow correct medical standards and that the app remains functional, accessible, and

up to date. They manage content changes, translations, and technical improvements to keep the system reliable for all other user groups.

2.3. Drawbacks of current system

The current landscape of first-aid resources in Ethiopia, comprising traditional instructional materials, general international mobile applications, and reliance on emergency services, presents several significant drawbacks that severely limit their effectiveness, particularly during critical, immediate-response scenarios.

2.3.1. Connectivity and Accessibility Limitations

The most significant barrier is the dependence of modern solutions on a consistent internet connection.

- **Reliance on Online Applications:** Most internationally available first-aid applications require an active internet connection (Wi-Fi or cellular data) to download content, load high-resolution images/videos, or integrate live mapping services. In many rural or remote regions of Ethiopia, internet coverage is intermittent or non-existent, rendering these tools completely useless when they are needed most.
- **Delayed Information Access:** Even in urban areas, a sudden lack of connectivity or slow data speeds can mean minutes of delay in retrieving critical step-by-step guidance, which can be fatal in time-sensitive emergencies like choking, severe bleeding, or cardiac arrest.

2.3.2. Linguistic and Cultural Gaps

The majority of high-quality digital first-aid resources are not tailored for the local population.

- **Language Barrier:** The overwhelming bulk of available, vetted digital first-aid content is published in English. This is inaccessible to the vast portion of the population who primarily speak Amharic or other local languages, leading to misinterpretation, confusion, and reduced confidence in administering aid.

- **Technical Jargon:** Even when translated, the instructions often use complex medical terminology or stiff, formal language that is difficult for a layperson to comprehend and execute quickly under pressure.
- **Inappropriate Scenarios:** International resources may reference equipment, infrastructure, or environmental factors (e.g., calling a specific foreign emergency number or using advanced medical kit items) that are not relevant or available in the local context.

2.3.3. Information Retrieval and Time Constraint

Existing methods are not optimized for rapid retrieval under duress.

- **Inefficient Search:** Traditional applications rely on keyword searches or hierarchical menus (e.g., scrolling through a list of conditions). In a panic, users struggle to articulate the exact medical term or navigate complex menus, leading to frustration and wasted time.
- **Lack of Natural Language Support:** No widely used system allows users to simply describe the situation in Amharic natural language (e.g., “*Something is stuck in a child’s throat*”) and immediately receive the corresponding protocol (Heimlich Maneuver), which significantly slows down the response time.
- **Static Media:** Printed materials (books, posters) are often outdated, not easily carried, and cannot adapt to user input or provide multimedia guidance, limiting their effectiveness in demonstrating procedures.

2.3.4. Limited Location Services for Emergencies

Current map solutions fail in the context of connectivity and localization for non-technical users.

- **Online Map Dependency:** Finding the nearest hospital or clinic requires an online map service (like Google Maps), which is disabled without internet, directly contravening the need for immediate, localized medical facility access.

- **Lack of Local Context Data:** While a user might find a major hospital online, many smaller, critical local health posts or clinics are often missing or poorly mapped, further complicating the journey to definitive care.

In summary, the current system is characterized by a reliance on internet connectivity, a pervasive language barrier to Amharic speakers, and inefficient information retrieval during high-stress situations. These drawbacks collectively highlight the unmet need for a dedicated, offline, and linguistically-tailored solution like the proposed system.

2.4. Literature Review

This section presents an overview of the research, studies, technologies, and existing systems related to mobile first-aid applications, digital health tools, offline information systems, and natural language processing for low-resource languages. The objective of the literature review is to understand the current state of knowledge in the field, identify what has already been achieved, determine the gaps that still exist, and establish a foundation for the methodology of the proposed system. By reviewing existing work, this project ensures that it is grounded in relevant theories and does not duplicate what has already been developed.

Several international organizations, including the International Federation of Red Cross and Red Crescent Societies (IFRC), the British Red Cross, and St John Ambulance, have developed mobile first-aid applications that offer structured step-by-step emergency guidance for common injuries. Research on these systems shows that mobile tools can significantly improve public preparedness and response during emergencies, especially when instructions are presented in simple, prioritized steps. Studies also highlight the importance of multimedia elements—such as images, diagrams, and short clips—to help users understand first-aid procedures more clearly.

Another area of literature relevant to this project focuses on offline access to digital health information. Many studies emphasize the need for offline-first systems in regions with limited connectivity, showing that preloaded content improves reliability and allows users to access guidance even during disasters or in rural areas. Existing mobile solutions often use device storage, downloadable content packs to ensure that essential information remains available without an internet connection.

In addition, research on natural language processing (NLP) in low-resource languages demonstrates the challenges and opportunities of developing intelligent language-based systems for languages such as Amharic. While multilingual language models and mobile-optimized NLP frameworks exist, the limited availability of annotated data and linguistic resources for Amharic requires careful adaptation and model optimization. Studies confirm that lightweight on-device NLP models can support tasks such as question answering and intent detection without requiring network connectivity, making them suitable for emergency or rural environments.

Furthermore, digital health research strongly supports the need for localized and culturally adapted first-aid information. Providing emergency guidance in a user's native language has been shown to increase comprehension, improve decision-making during emergencies, and build trust in digital health tools. Several projects across Africa and Asia emphasize that local-language content plays a critical role in bridging health information gaps in communities where English is not widely spoken.

From the reviewed literature, it is clear that although first-aid mobile systems and offline health tools exist globally, there remain major gaps in local-language accessibility, offline availability, and the integration of intelligent features for languages like Amharic. These gaps justify the development of the proposed system, which aims to combine offline first-aid guidance, Amharic-language NLP capability, and offline mapping features into a fully accessible and locally relevant mobile application.

This literature review thus supports the direction of the project by demonstrating the importance of mobile first-aid systems, validating the need for offline and localized content, and highlighting the research gaps that the proposed system intends to address.

Chapter 3 : Proposed System

3.1 Overview

The proposed system, **Offline Amharic First Aid Advisor Using NLP and Offline Maps**, will be a mobile application designed to provide reliable, accurate, and offline-accessible first aid guidance to Ethiopian users. The system integrates three major components:

1. **Offline Amharic NLP Engine** – interprets user queries typed or spoken in natural Amharic and classifies the emergency type (e.g., bleeding, choking, burns).
2. **First Aid Knowledge Base Module** – a local SQLite database containing step-by-step instructions, images, and audio guidance for common emergencies.
3. **Offline Map & Location Module** – provides nearby hospital listings and basic navigation using preloaded map tiles.

The primary purpose of the system is to deliver **fast, localized, and understandable** first aid assistance during emergency situations—especially in environments where internet connectivity is absent or weak. The system emphasizes:

- Offline operation
- Amharic-language interaction
- Simple and clear instructions
- Lightweight mobile performance

The overall architecture follows a **mobile-first offline-first model**, ensuring usability on low-end Android devices common in Ethiopia.

3.2 Functional Requirements

The functional requirements describe the specific function and behaviors that the **Offline Amharic First Aid Advisor Using NLP and Offline Maps** system must provide. These

requirements define what the system should do to meet user needs in both emergency and non emergency situations.

3.2.1 First Aid Guidance Module

- **FR-01:** The system shall allow users to browse emergency categories (bleeding, burns, choking, fractures, etc.).
- **FR-02:** The system shall display step-by-step first aid instructions for each emergency.
- **FR-03:** The system shall provide images and audio explanations for every guide.
- **FR-04:** The system shall allow users to search emergencies manually using keywords.
- **FR-05:** The system shall allow users to bookmark or favorite guides.

3.2.2 Amharic NLP Module

- **FR-06:** The system shall accept Amharic text-based queries.
- **FR-07:** The system shall accept Amharic speech input offline.
- **FR-08:** The NLP engine shall classify the user's emergency intent (e.g., “ልጅ ተከሰተ” → “Bleeding”).
- **FR-09:** The system shall display relevant first aid guidance based on NLP output.
- **FR-10:** The system shall suggest the closest matching emergencies if the query is unclear.

3.2.3 Offline Map & Hospital Locator Module

- **FR-11:** The system shall display an offline map using locally stored tiles.
- **FR-12:** The system shall show markers for hospitals, clinics, and health posts.
- **FR-13:** The system shall retrieve the user's GPS location without internet.
- **FR-14:** The system shall calculate and display “Hospitals Near Me.”
- **FR-15:** The system shall show a basic route line from user location to the selected hospital.

3.2.4 Utility & Support Tools

- **FR-16:** The system shall include a First Aid Kit checklist.

- **FR-17:** The system shall provide an emergency call shortcut (907/939).
- **FR-18:** The system shall operate fully offline after installation.

3.3 Non-Functional Requirements

Non-functional requirements describe how the system should perform rather than what it should do. These requirements focus on the quality attributes of the **Offline Amharic First Aid Advisor Using NLP and Offline Maps** application such as performance, reliability, usability, security, scalability and maintainability.

Performance

- **NFR-01:** App launch time shall not exceed 5 seconds.
- **NFR-02:** NLP classification shall respond within 1–3 seconds.
- **NFR-03:** Map rendering shall complete in under 5 seconds.

Reliability

- **NFR-04:** The system shall work without internet at all times.
- **NFR-05:** Local data must remain intact even after unexpected shutdowns.
- **NFR-06:** The system shall not crash during rapid navigation between modules.

Usability

- **NFR-07:** The UI shall use simple Amharic terms for laypersons.
- **NFR-08:** Emergency steps shall be presented in large fonts and icons.
- **NFR-09:** The system shall support rural, low-literacy users through audio and images.

Security

- **NFR-10:** No personal data shall be collected or transmitted.
- **NFR-11:** Location data shall remain local and never stored externally.

Maintainability

- **NFR-12:** System modules (NLP, Maps, Database) must be decoupled.

- **NFR-13:** Codebase must follow clean architecture for future updates.

Scalability

- **NFR-14:** The system shall allow adding new emergency types in future versions.
- **NFR-15:** The NLP engine shall support future language extensions.

3.4 System model

The System Model represents how the Offline Amharic First Aid App is structured, how its components interact, and how it behaves when processing user actions. It explains the system from three different perspectives:

1. Conceptual Model

The conceptual model defines the **high-level** structure and the **major components** of the system without going into implementation details. It answers what the system does, what modules exist and how users interact with them.

Main Components (Concept Level)

1. User Interface (UI Module)

- Displays categories, instructions, NLP button, search, maps, and calling options.

2. NLP Processing Module

- Accepts Amharic voice/text.
- Identifies the emergency (e.g., choking, bleeding).
- Sends recognized intent to instruction retriever.

3. Instruction Retrieval Module

- Maps the emergency type to the correct offline first-aid guide.
- Loads text, images, and audio instructions.

4. Offline Database (First Aid Knowledge Base)

- Contains all emergencies, step-by-step instructions, images, audio guides, and categories.

5. Offline Map & Location Module

- Shows offline map tiles.
- Uses GPS to find user location.
- Locates nearest hospitals/clinics.

6. Emergency Communication Module

- Integrates with device phone dialer.
- Lets users call clinics or emergency numbers.

7. User (Actor)

- Regular user
- Learner
- Parent in emergency
- Bystander
- Health volunteer

Conceptual Workflow

User → UI → (Category / NLP / Search / Map / Calling) → Instruction Database → Action/Guidance

User → GPS → Map → Clinic

User → Phone → Emergency contact

2. Logical Model

The logical model describes how the system is organized internally—how data flows between modules, how processes are triggered, and how components interact logically (not physically).

It answers the questions How does the system work step by step, how do modules communicate and which component depends on the other.

Logical Architecture Components

1. Presentation Layer (UI Layer)

- Home screen
- Category list

- Learning mode page
- NLP speech input interface
- Search bar
- Offline map interface
- Emergency contacts page

2. Application Logic Layer

- NLP processor (intent detection)
- Emergency classifier
- Instruction manager
- Bookmark handler
- Audio playback controller
- Clinic locator logic
- Call initiation logic

3. Data Layer

- SQLite database storing:
 - Emergencies
 - Steps
 - Illustrations
 - Audio files
 - Clinic info
 - Map tiles

Logical Data Flow

1. User selects an option (category, NLP, search, map, call).
2. Input is passed to the logic layer (NLP Engine or Category Handler).
3. Logic layer queries the data layer.
4. Data is returned (instructions, map points, clinic details).
5. UI presents results.

6. Optional: user performs actions (call, listen audio, navigate).

3. Physical Model

The physical model describes the real-world deployment environment and hardware/software components. It answers Where does the application run, what hardware does it require and where is the database stored physically.

Physical Components

1. Android Smartphone (Device Hardware)

- CPU & RAM for NLP inference
- Microphone for speech input
- GPS chip
- Storage for offline data
- Phone dialer

2. Application Storage (Local)

- SQLite database file
- NLP model file
- Offline map tiles
- Audio guide files
- Image files

3. Application Software (React Native or similar)

- UI built with React Native
- Offline NLP via TensorFlow Lite
- Local storage via SQLite

Deployment Diagram Summary

Mobile Device (App + Local Storage + Sensors)

Runs the entire system offline—no cloud or server required

3.4.1. Scenario

Scenario 1: Category-Based Browsing for First Aid Guidance

Actor : Regular User / Any App User

Goal : To find first aid instructions by selecting from the list of categories.

Preconditions:

- The app is installed and functioning offline.
- First aid categories and instructions are loaded in the local database.

Main Flow (Steps):

1. User opens the mobile application.
2. The home screen displays a list of first aid categories (Bleeding, Burns, Choking, Fractures, etc.).
3. User scrolls through the categories.
4. User identifies the category relevant to their need.
5. User taps the chosen category.
6. The system loads and displays the related step-by-step instructions.
7. User reviews the steps and may view images or play audio guidance.

Postconditions:

- The user has accessed and understood the correct first aid instructions.

Scenario 2: Learning First Aid Topics for Educational Purposes

Actor : Learner / Student / Health Volunteer

Goal : To study first aid topics in a non-emergency situation.

Preconditions:

- The app is installed.
- Offline first aid database is available.
- No emergency situation is occurring.

Main Flow (Steps):

1. User launches the app.
2. User selects “First Aid Topics” or a specific category to learn.
3. The system displays detailed descriptions, steps, and illustrations.
4. User reads the content thoroughly.
5. User listens to optional audio explanations.
6. User bookmarks the topic for future study.

Postconditions:

- The user gains knowledge about first aid procedures and becomes more prepared.

Scenario 3: NLP-Based Emergency Assistance (Child Choking)

Actor : Parent / Guardian

Goal : To receive rapid first aid guidance through Amharic speech input.

Preconditions:

- The NLP module is installed and functional offline.
- Microphone permissions are enabled.
- The user is experiencing an emergency.

Main Flow (Steps):

1. User opens the app in panic.
2. User taps the “Ask in Amharic” voice button.
3. User says: “ልጄ እንቀው እንዴት ልርዳው?”
4. NLP engine processes the input offline.
5. System identifies the intent as **Choking (Child)**.
6. System immediately displays the choking first aid guide.
7. User follows the step-by-step instructions.
8. User may play audio for hands-free guidance.
9. User opens the offline map to find a clinic if needed.

Postconditions:

- The user receives correct and immediate guidance for the emergency.

Scenario 4: Locating the Nearest Hospital Using Offline Maps

Actor : Any User

Goal : To find the nearest hospital or clinic using offline map navigation.

Preconditions:

- GPS is enabled.
- Offline map tiles and hospital database are available.

Main Flow (Steps):

1. User opens the “Nearby Hospitals” section.
2. App retrieves the user's GPS coordinates offline.
3. App loads the offline map.
4. App displays nearby hospitals/clinics as markers.
5. User selects a health facility.
6. App displays a simple offline route or direction line.
7. User follows the directions to reach the clinic.

Postconditions:

- User identifies and navigates to a nearby medical facility.

Scenario 5: Calling and Contacting Local Clinics for Emergency Assistance

Actor : Any User / Bystander / Family Member

Goal : To quickly contact a nearby clinic or emergency service for immediate medical assistance.

Preconditions:

- The app is installed and functioning offline.
- Offline clinic database with phone numbers is available.

- User's device supports phone calls.

Main Flow (Steps):

1. User opens the application.
2. User navigates to the “Emergency Contacts” or “Nearby Clinics” section.
3. App displays a list of clinics, hospitals, and emergency service numbers (907/939).
4. User selects the clinic or emergency service they want to call.
5. App displays the facility’s phone number, name, and brief details.
6. User taps the “Call Now” button.
7. System opens the phone dialer automatically with the number pre-filled.
8. User initiates the call and speaks with medical personnel for instructions or to request help.

Postconditions:

- The user successfully contacts a clinic or emergency service.
- Emergency responders are informed and help is on the way, or guidance is provided.

3.4.2. Use case model

The use case diagram presents a high-level visualization of the interactions between users and the Offline Amharic First Aid Application. A generalized “User” actor is used to represent different user roles, including general users, learners, parents, and bystanders. These roles share common interactions with the system, especially in accessing first aid information and emergency assistance. It highlights how users receive first aid guidance, learn educational content, locate clinics offline, and contact emergency services.

Actors

1. General User

- Regular users who interact with the system to browse first aid categories, view instructions, search topics, and use basic app features.

2. Emergency User / Bystander / Parent

- A user who interacts with the system in urgent situations to receive immediate guidance through voice (NLP), locate clinics, or contact emergency services.

3. Learner / Student / Health Volunteer

- Users who engage with the system for educational purposes, reading first aid topics, bookmarking lessons, and studying procedures.

4. System Actors

- These are not humans but internal or device-based components that play important roles in use cases:
 - **NLP Engine:** Processes Amharic voice input offline and identifies the emergency intent.
 - **GPS Module:** Provides the device's current location for offline maps.
 - **Phone Dialer:** Enables users to call clinics or emergency services.

Use Case Descriptions

1. General User Use Cases

- **Browse First Aid Categories :** Allows users to scroll through and select categories such as Bleeding, Burns, Fractures, Choking, etc., and access relevant first aid instructions.
- **View Step-by-Step Instructions :** Users can read detailed instructions, view images, and optionally play audio guidance for any selected first aid topic.
- **Search First Aid Content :** Users can search for specific injuries, symptoms, or emergencies using the built-in search feature.
- **View Offline First Aid Resources :** Users access all content offline, including images, audio, categories, and instructions.

2. Learner / Student Use Cases

- **Learn First Aid Topics :** Learners study procedures in a non-emergency context, viewing detailed explanations, illustrations, and examples.
- **Bookmark Topics :** Learners can save specific procedures or lessons for future reference.
- **Review Saved Content :** Users return to previously saved or bookmarked first aid topics.

3. Emergency User Use Cases

- **Invoke NLP Emergency Assistance :** Users speak Amharic emergency queries like “ልጄ ለጎብኙ” to get instant first aid instructions.
- **Submit Voice Input (Speak Query) :** The app receives voice input for processing during emergencies.
- **Receive Immediate Emergency Instructions :** The system displays relevant emergency steps instantly after identifying the intent.
- **Play Hands-Free Audio Guidance :** During an emergency, users may choose to listen to the instructions instead of reading them.
- **Locate Nearby Clinics (Offline) :** Users find clinics or hospitals using GPS and offline maps during an emergency.

- **Contact Emergency Services (907/939)** : Users quickly dial emergency responders for assistance.

4. System Actor Use Cases

- **NLP Engine Use Cases**
 - Process Speech Input: Receives speech and converts it to text.
 - Identify Intent: Determines what emergency the user is describing.
 - Return Emergency Type: Sends emergency type (e.g., choking, bleeding) back to the UI.
- **GPS Module Use Cases**
 - Retrieve User Location: Gets offline coordinates.
 - Provide Nearby Facilities: Supplies location data to the map component.
- **Phone Dialer Use Cases**
 - Dial Emergency Numbers: Opens the phone dialer with emergency numbers pre-filled.
 - Dial Clinic Numbers: Enables quick calling of nearby clinics.

Use Case Diagram

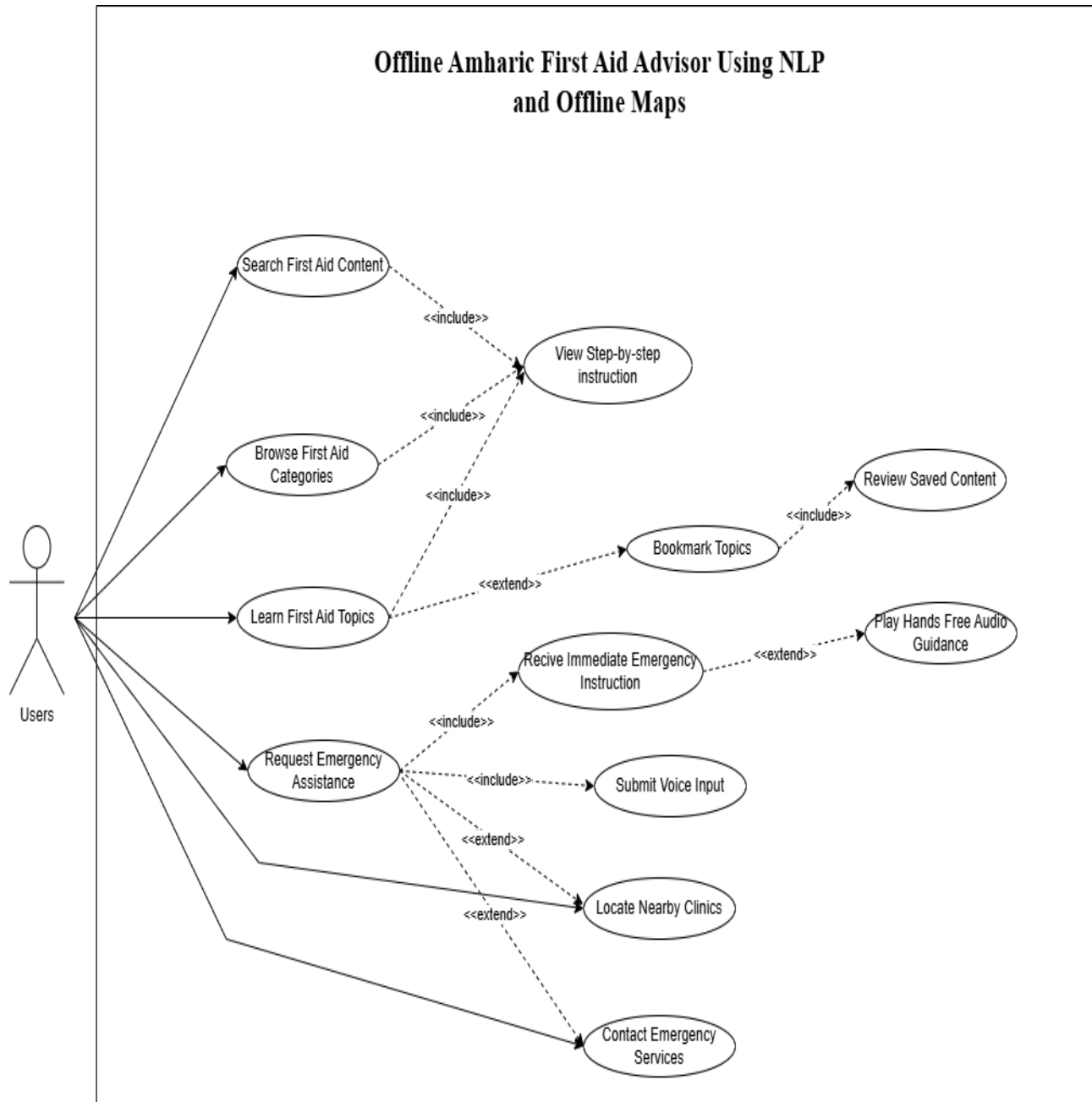


Figure 1 Use Case Diagram

3.5.1 Data Dictionary

The Data Dictionary provides detailed definitions of the data elements used within the Offline Amharic First Aid Mobile Application. It includes the names, data types, formats, possible

values, and descriptions of each data element to ensure clarity and consistency throughout the system.

1. User Session Data Elements

Element Name	Data Type	Format	Possible Values	Description
sessionId	String	UUID	e.g., a1b2c3d4-e5f6-7890-1234-567890abcdef	Unique identifier for the current app session
currentLocation	Object	JSON	{"latitude": 9.145, "longitude": 40.4897, "accuracy": 15}	User's current GPS coordinates
selectedCategory	String	Text	"Bleeding", "Burns", "Choking", "Fractures", etc.	Currently selected first aid category
activeProcedure	String	UUID	UUID format	Currently displayed first aid procedure ID
audioPlaying	Boolean	true/false	true, false	Whether audio guidance is currently playing

Table 4 User Session Data Elements

2. First Aid Category Data Elements

Element Name	Data Type	Format	Possible Values	Description
categoryId	String	UUID	UUID format	Unique identifier for each category
categoryName	String	Text	"እርዳታ ለመጀመሪያ ጊዜ", "ቆሻሻ", "ማቃጠል", etc.	Amharic name of the category
categoryNameEn	String	Text	"First Aid", "Bleeding", "Burns", etc.	English name of the category (for development)
iconName	String	Text	"bleeding_icon", "burns_icon", "choking_icon"	Reference to icon file in assets
description	String	Text	Short description of the category	Brief explanation of what the category covers
emergencyLevel	Integer	Number	1–5 (1=low urgency, 5=critical)	Urgency level for triage display
procedureCount	Integer	Number	e.g., 12	Number of procedures in this category

Table 5 First Aid Category Data Elements

3. First Aid Procedure Data Elements

Element Name	Data Type	Format	Possible Values	Description
procedureId	String	UUID	UUID format	Unique identifier for each procedure
categoryId	String	UUID	UUID format	Foreign key to category
titleAm	String	Text	"የደም ፍሳሽን እንዴት መቆጣጠር እንደሚቻል"	Procedure title in Amharic
titleEn	String	Text	"How to Control Bleeding"	Procedure title in English
emergencyType	String	Enum	"Medical", "Trauma", "Environmental", "Pediatric"	Type of emergency
severityLevel	String	Enum	"Mild", "Moderate", "Severe", "Critical"	Severity of the condition
estimatedTime	Integer	Number	e.g., 300 (seconds)	Estimated time to complete procedure
equipmentNeeded	Array	JSON	["gloves", "bandage", "clean cloth"]	List of required equipment

warningAm	String	Text	"ማስታወሻ: ይህ እርዳታ የህክምና ምክር አይደለም"	Warning notice in Amharic
warningEn	String	Text	"Note: This is not medical advice"	Warning notice in English

Table 6 First Aid Procedure Data Elements

4. Procedure Step Data Elements

Element Name	Data Type	Format	Possible Values	Description
stepId	String	UUID	UUID format	Unique identifier for each step
procedureId	String	UUID	UUID format	Foreign key to procedure
stepNumber	Integer	Number	1, 2, 3, ...	Sequential step number
instructionAm	String	Text	"ጥቂት ጊዜ ድንጋይ እስቀምጠው ይደዙ"	Step instruction in Amharic
instructionEn	String	Text	"Apply pressure with a clean cloth"	Step instruction in English

imageRef	String	Text	"bleeding_step1.jpg", "choking_diagram.png"	Reference to image file
audioRef	String	Text	"step1_audio.mp3", "bleeding_guide.aac"	Reference to audio file
durationEstimate	Integer	Number	e.g., 30 (seconds)	Estimated time for this step
criticalFlag	Boolean	true/false	true, false	Whether step is critical (must not skip)
cautionAm	String	Text	"ያስተውሉ: በጣም ጠንካራ አይደለም"	Caution note in Amharic

Table 7 Procedure Step Data Elements

5. NLP Processing Data Elements

Element Name	Data Type	Format	Possible Values	Description
queryId	String	UUID	UUID format	Unique identifier for each query
inputText	String	Text	"ልጄ አነቀው", "ደም ይፈሳል"	User's Amharic text input

inputAudio	Blob	Binary	Audio recording data	Raw audio data from microphone
processedText	String	Text	"ልጅ ለነቀው"	Cleaned/processed text
detectedIntent	String	Enum	"CHOKING", "BLEEDING", "BURNS", "FRACTURE"	NLP-detected emergency type
timestamp	DateTime	ISO 8601	2024-03-15T10:30:00Z	When query was processed
matchedProcedureId	String	UUID	UUID format	Procedure ID matched by NLP

Table 8 NLP Processing Data Elements

6. Medical Facility Data Elements

Element Name	Data Type	Format	Possible Values	Description
facilityId	String	UUID	UUID format	Unique identifier for facility
nameAm	String	Text	"ቶረኖ ሆስፒታል"	Facility name in Amharic

nameEn	String	Text	"Torino Hospital"	Facility name in English
facilityType	String	Enum	"Hospital", "Clinic", "Health Center", "Pharmacy"	Type of medical facility
latitude	Float	Number	9.0320, 8.9802, etc.	GPS latitude coordinate
longitude	Float	Number	38.7465, 38.7890, etc.	GPS longitude coordinate
phoneNumber	String	Phone	"+251911223344", "907", "939"	Contact phone number
addressAm	String	Text	"አዲስ አበባ, ቦሌ"	Address in Amharic
addressEn	String	Text	"Addis Ababa, Bole"	Address in English
operatingHours	String	Text	"24/7", "8AM-10PM"	Operating hours
services	Array	JSON	["Emergency", "Pediatrics", "Surgery"]	Services offered

Table 9 Medical Facility Data Elements

Data Element Descriptions in Detail

User Session Elements

- **sessionId:** A unique identifier generated when the app starts, used to track a single usage session without storing personal data.
- **currentLocation:** Object containing latitude, longitude, and accuracy (in meters) from GPS. Used for map features and locating nearby clinics.
- **selectedCategory:** The currently active category ID from user selection, used to filter and display relevant procedures.

First Aid Category Elements

- **categoryId:** Primary key for categories, referenced by procedures.
- **emergencyLevel:** Numerical value (1-5) where 1 represents non-urgent categories (like "Basic First Aid") and 5 represents critical emergencies (like "Cardiac Arrest").
- **iconName:** Reference to the icon file stored in the app's assets, displayed in the category selection screen.

First Aid Procedure Elements

- **procedureId:** Primary key for procedures, referenced by steps, bookmarks, and search results.
- **estimatedTime:** In seconds, helps users understand time commitment before starting.
- **equipmentNeeded:** JSON array of strings, displayed to help users gather necessary items before starting the procedure.
- **warningAm/En:** Legal disclaimer in both languages to emphasize that this is first aid guidance, not professional medical advice.

Procedure Step Elements

- **stepNumber:** Ensures steps are displayed in correct order (1, 2, 3...).
- **imageRef/audioRef:** File references stored locally in the app's assets folder.
- **criticalFlag:** If true, this step cannot be skipped and is highlighted in the UI.

- **cautionAm:** Additional safety warnings specific to that step in Amharic.\

NLP Processing Elements

- **detectedIntent:** Output from the TensorFlow Lite model, matching user's spoken/written query to an emergency type.
- **matchedProcedureId:** The procedure ID that the NLP system determines is most relevant to the user's query.

Medical Facility Elements

- **facilityType:** Categorizes facilities for filtering (e.g., show only hospitals, not pharmacies).
- **operatingHours:** Free-text field as operating hours vary widely in Ethiopia.
- **services:** JSON array allowing filtering by available services (e.g., show facilities with "Pediatrics").

3.5.2 Class Diagram

This diagram illustrates the main classes of the proposed system, including the mobile application, Amharic NLP engine, first aid management, offline maps, and utility modules. It shows key attributes, methods, and the relationships between classes, providing a clear overview of the system's structure and functionality.

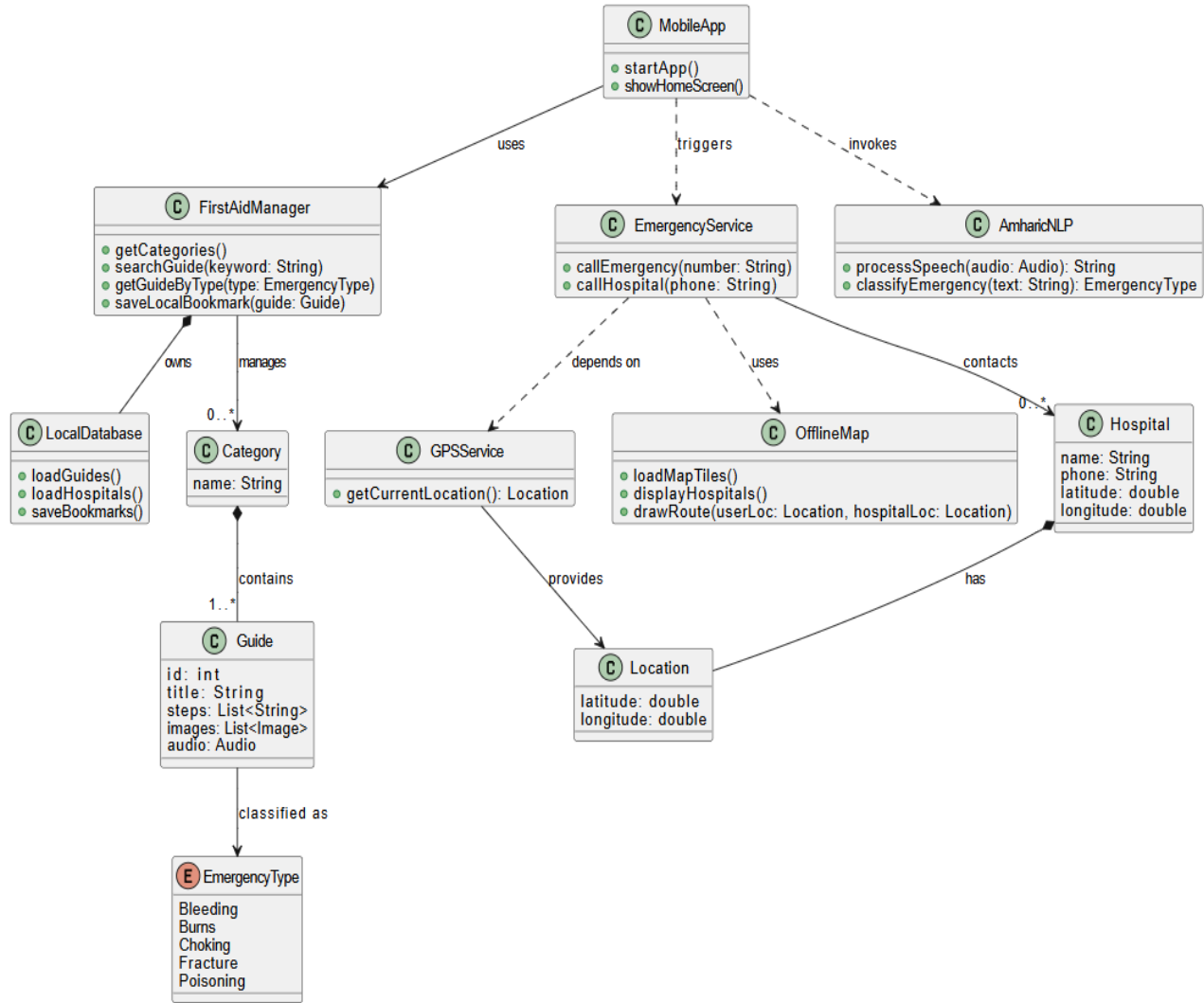


Figure 2 Class Diagram

3.5.3 Dynamic Model

The Dynamic Model describes how the Offline Amharic First Aid Advisor Using NLP and Offline Maps system behaves over time. It focuses on the changes in system states, the events that trigger these changes, and the actions performed by the system in response. This model helps to understand how different components of the system interact dynamically during normal

and emergency operations.

The dynamic behavior of the system is represented using state transitions, events and actions that occur when a user interacts with the application.

System States

The system moves through various states during its operation:

- Idle State – The system is waiting for user interaction.
- Home Screen Active – The main interface is displayed.
- Input Received – The system captures user text or voice input.
- Processing State – The NLP engine or database processes the request.
- Guide Display State – First aid instructions are shown to the user.
- Map Display State – Offline map and nearby hospitals are displayed.
- Emergency Call State – The system prepares and launches the phone dialer.
- Paused State – The system is temporarily inactive due to interruption.
- Exit State – The application is closed.

Events and Transitions

The following events trigger transitions between states:

Current State	Event	Next State	Action Performed
Idle	User opens the app	Home Screen Active	Load main interface
Home Screen Active	User selects a category	Processing State	Fetch data from local database
Home Screen Active	User speaks or types query	Input Received	Capture voice/text input
Input Received	System starts NLP	Processing State	Classify emergency type
Processing State	Classification complete	Guide Display State	Show first aid steps

Guide Display State	User opens map	Map Display State	Load offline map tiles
Map Display State	User selects a hospital	Map Display State	Draw route line
Any Active State	User taps “Call”	Emergency Call State	Launch phone dialer
Any State	App minimized	Paused State	Save current session
Paused State	User returns	Previous State	Restore session
Any State	User exits app	Exit State	Close application

Table 10 Events and Transitions

Dynamic Behavior Summary

The system continuously reacts to user actions and system events. It receives user input, processes it locally using the NLP engine and database, and responds by displaying appropriate first aid guidance or navigation support. All operations are designed to work fully offline to ensure reliability during emergencies.

3.5.4. Sequence diagram

This is a diagram that shows the sequence or order of the messages or communications between the objects or components of the proposed system or solution, such as the sender, receiver, message, or time of a system operation or function.

3.5.4.1. Sequence Diagram Category-Based First Aid Browsing

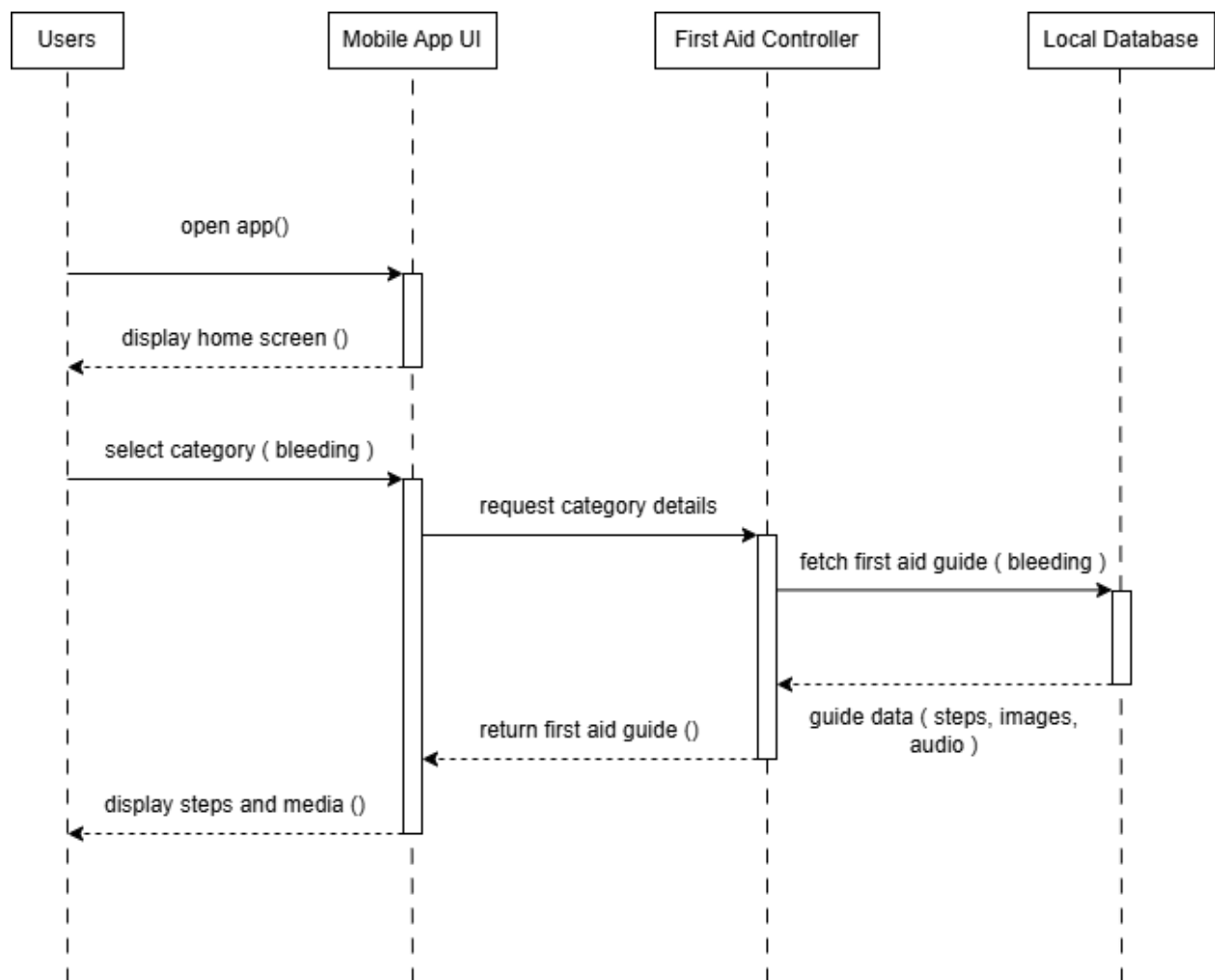


Figure 3 Sequence Diagram Category-Based First Aid Browsing

3.5.4.2. Sequence Diagram NLP Emergency Query (Amharic)

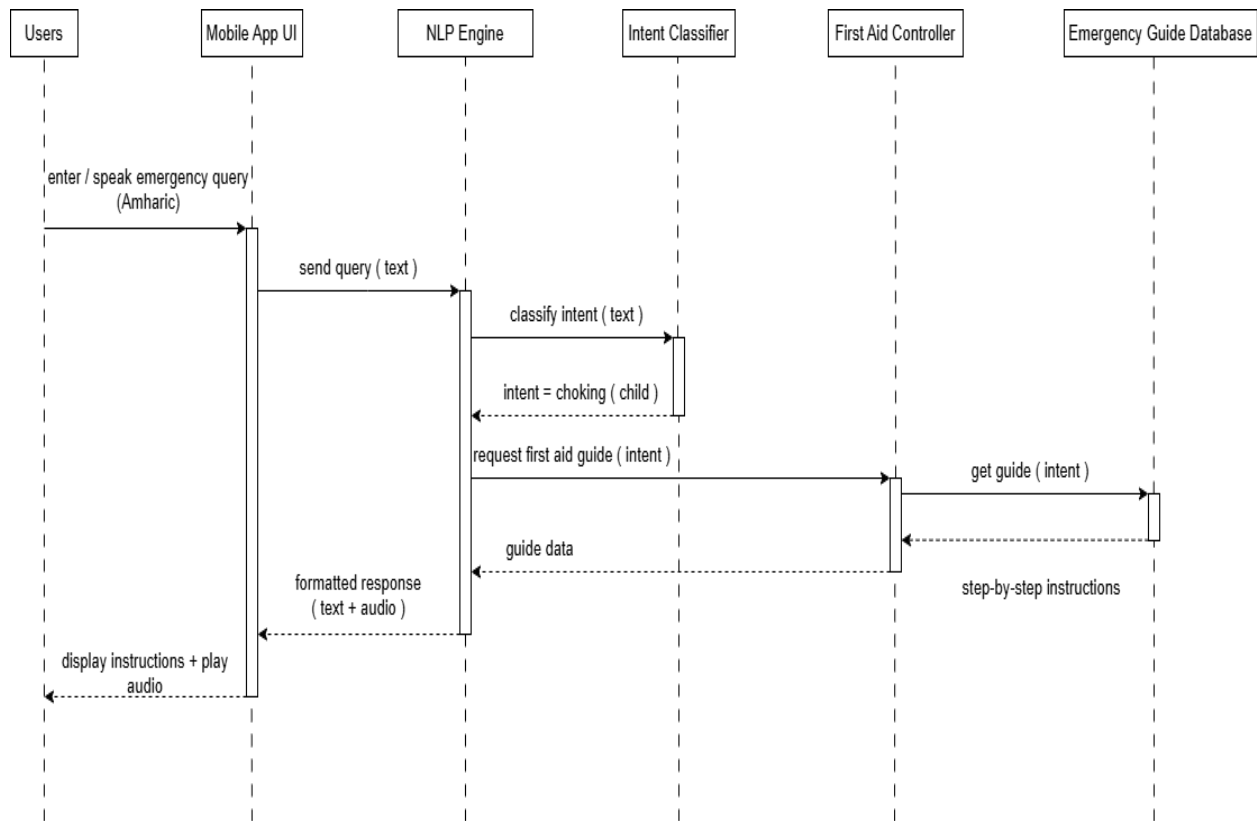


Figure 4 Sequence Diagram NLP Emergency Query (Amharic)

3.5.4.3. Sequence Diagram Offline Hospital Locator

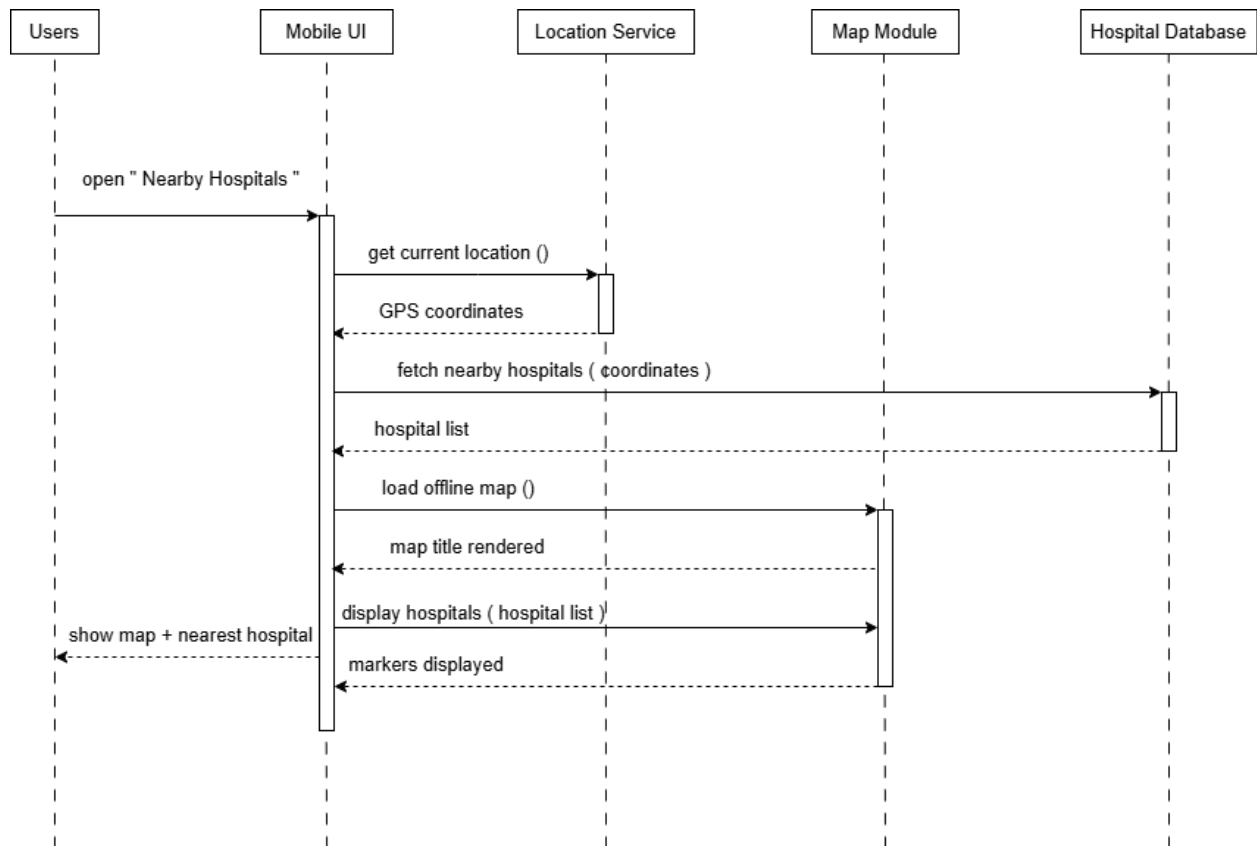


Figure 5 Sequence Diagram Offline Hospital Locator

3.5.4.4. Sequence Diagram Emergency Call Shortcut

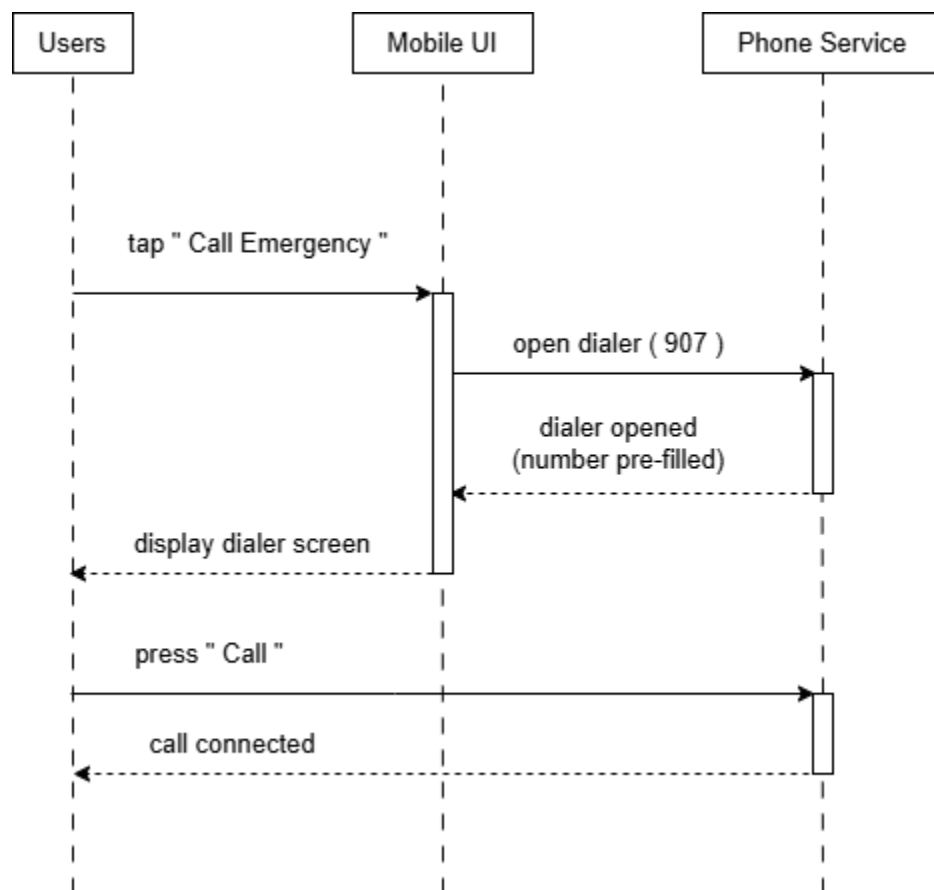


Figure 6 Sequence Diagram Emergency Call Shortcut

3.5.5. Activity diagram

1. Activity Diagram for Category-Based Browsing

- This diagram shows the normal user flow where a user browses first aid categories from the home screen.
- It illustrates how the system displays available categories, how the user scrolls, and selects a relevant category.

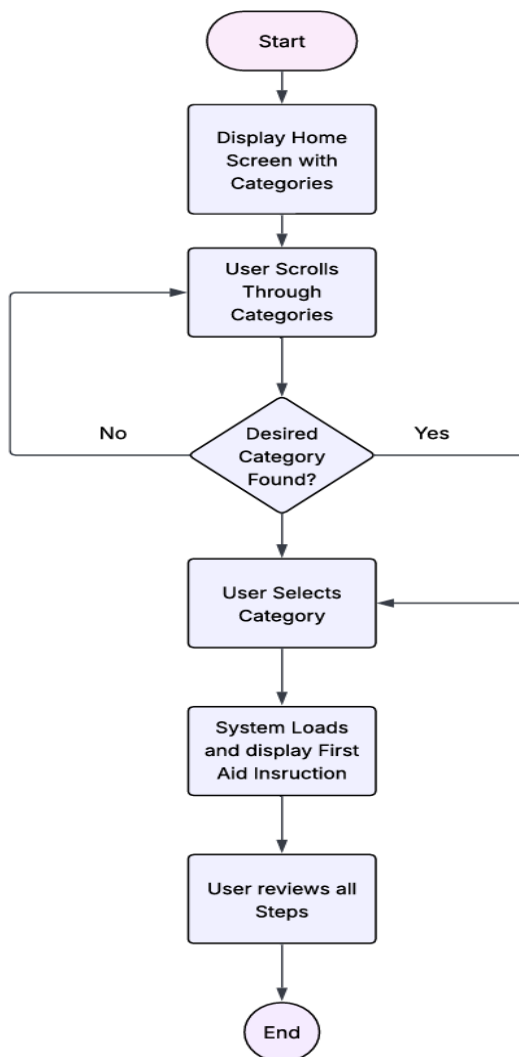


Figure 7 Activity Diagram for Category-Based Browsing

2. Activity diagram for Learning First Aid Topics

- This diagram models the educational flow where users study first aid topics without an emergency.
- It shows how users navigate to the learning section, select a topic, read content, and optionally play audio explanations.
- It includes the decision to bookmark a topic for future reference.

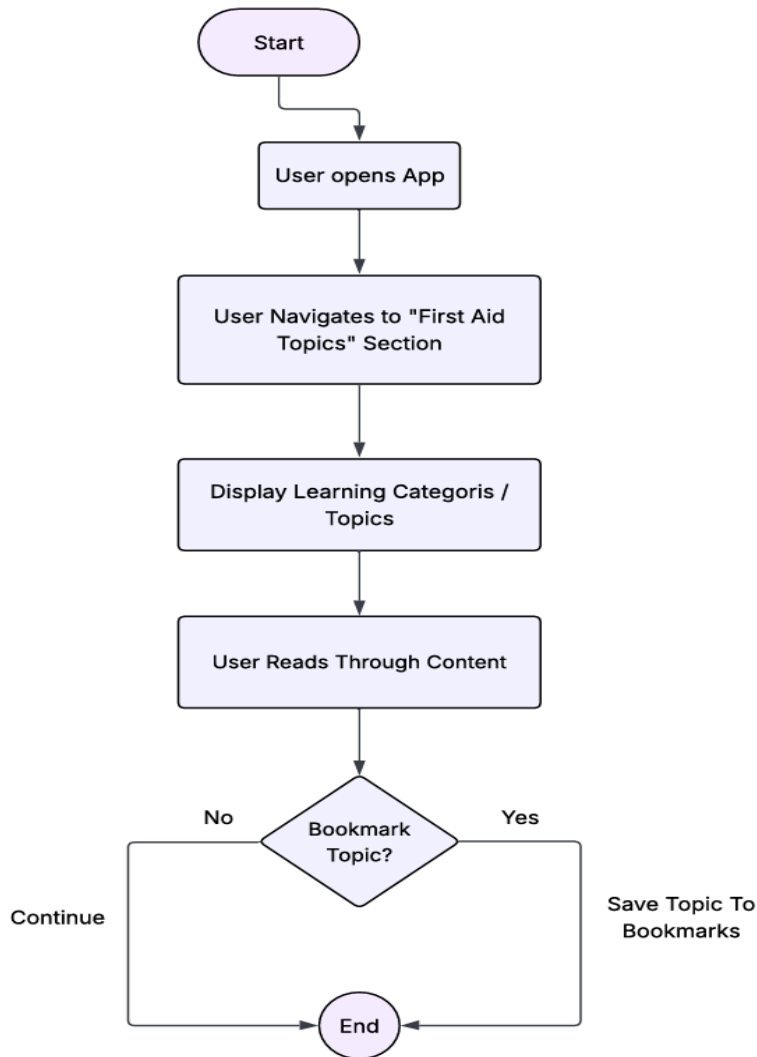


Figure 8 Activity diagram for Learning First Aid Topics

3. Activity diagram for NLP-Based Emergency Assistance

- This diagram represents the emergency workflow where a user relies on voice-based NLP assistance during a critical event. It shows how the system receives speech, performs offline intent recognition, and displays emergency instructions.

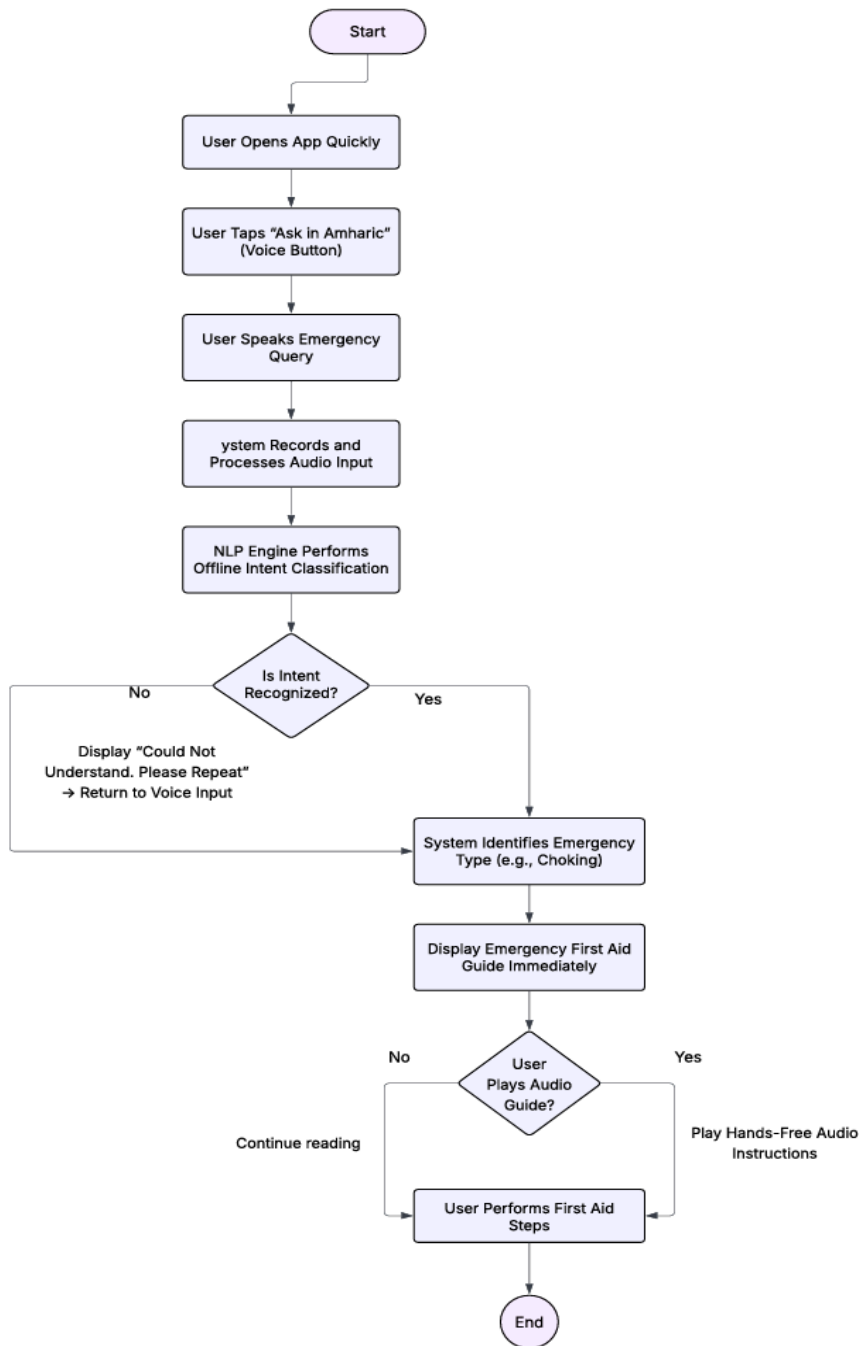


Figure 9 Activity diagram for NLP-Based Emergency Assistance

4. Activity diagram for Locating the Nearest Hospital Using Offline Map

- This diagram describes how a user finds nearby clinics or hospitals using offline GPS and map data.
- It shows key steps including GPS retrieval, map loading, facility selection, and route visualization.
- It emphasizes offline functionality, all navigation happens without internet access.

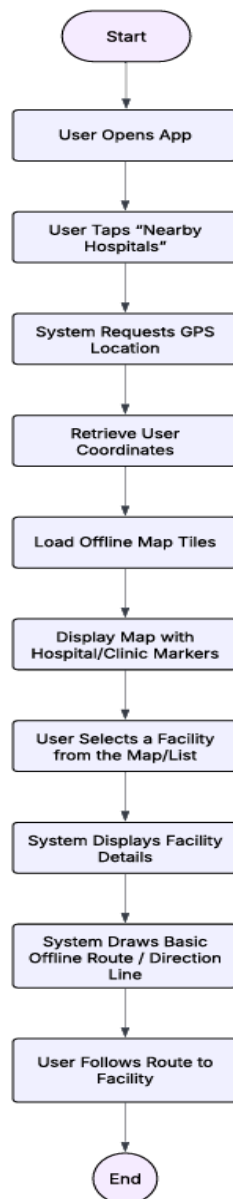


Figure 10 Activity diagram for Locating the Nearest Hospital Using Offline Map

3.5.6. State chart diagram

State Diagram 1: First Aid Procedure Session State

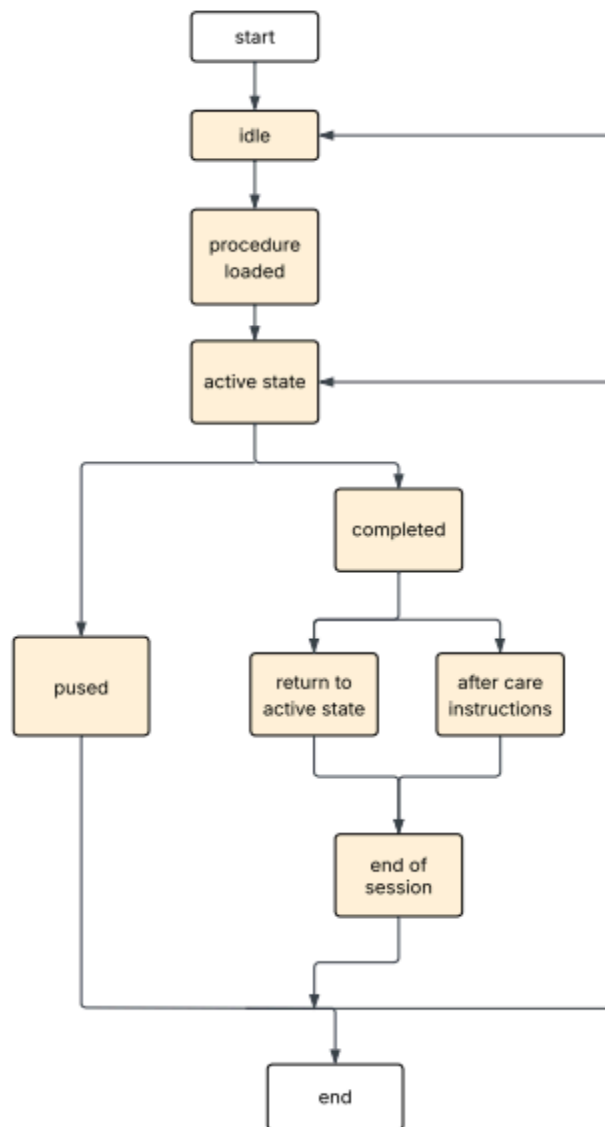


Figure 11 First Aid Procedure Session State

State Descriptions

Idle: No procedure is active; user is browsing or searching

Procedure Loaded: Emergency procedure loaded but not yet started

Step Active: User actively following step-by-step instructions

Paused: Procedure temporarily halted (user interrupted)

Completed: All steps finished; aftercare instructions displayed

State Diagram 2: NLP Query Processing State

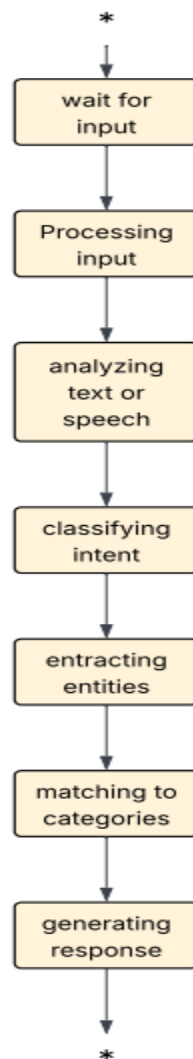


Figure 12 NLP Query Processing State

State Diagram 3: Offline Map Session State

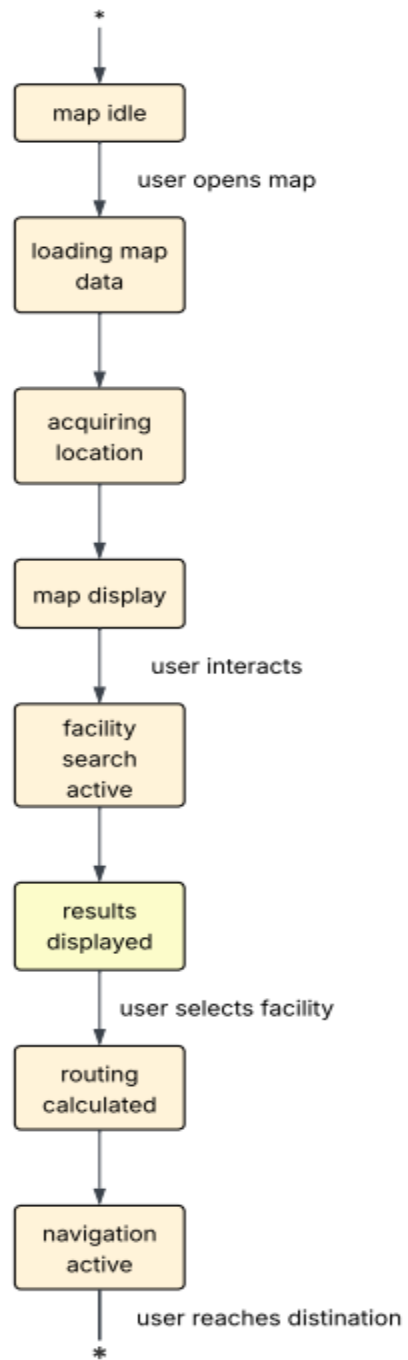


Figure 13 Offline Map Session State

Chapter 4: System design

4.1 Overview

This chapter will present the system design of the proposed **Offline Amharic First Aid Advisor Using Natural Language Processing and Offline Maps**. The system design describes the overall structure of the application, its major components and how these components interact to deliver offline first aid guidance and location-based services.

The proposed system will follow a mobile-first and offline-first architecture, ensuring that all core functionalities operate without internet connectivity. The design integrates multiple subsystems, including the user interface, Amharic NLP processing module, first aid content management, offline mapping services and local data storage.

At a high level, the system consists of the following main components:

- **User Interface (UI) Module:** Provides an intuitive and user-friendly interface that allows users to interact with the system using Amharic text or voice inputs and navigate first aid content easily.
- **Amharic NLP Module:** Processes user queries written or spoken in Amharic, identifies the type of emergency, and maps it to appropriate first aid instructions.
- **First Aid Knowledge Base:** Stores structured first aid instructions, multimedia content (text, images, audio), and emergency categories in an offline local database.
- **Offline Map and Hospital Locator Module:** Enables users to locate nearby hospitals, clinics, and health centers using preloaded offline map data.
- **Local Database Management Module:** Uses an embedded database (SQLite) to store first aid content, hospital information, and system metadata
- **System Control and Integration Layer:** Coordinates communication between all subsystems and ensures smooth data flow and execution.

4.2 Purpose of the System Design

The purpose of the system design will be to provide a clear and structured blueprint for implementing the proposed system in a reliable, efficient and scalable manner. Specifically, the system design aims to:

Meet Functional Requirements

The design ensures that all functional requirements such as Amharic based emergency query handling, step-by-step first aid guidance, offline hospital location, and multimedia support are fully supported by well defined system components.

Satisfy Non Functional Requirements

The architecture emphasizes performance, reliability, usability, security and maintainability. Offline processing minimizes dependency on external services, improving reliability in low connectivity areas.

Support Offline Operation

A key design objective is complete offline functionality. The system design ensures that NLP processing, first aid guidance, and mapping services rely on local models and databases rather than cloud-based services.

Facilitate Modular Development and Testing

By separating the system into independent modules (UI, NLP, database, mapping), the design simplifies development, debugging, testing and future enhancement.

Ensure Ease of Maintenance and Future Expansion

The design allows new emergency categories, languages, or features (such as additional Ethiopian languages or online synchronization) to be added without major architectural changes.

Align with System and Object Models

The system design complements the system models, class diagrams and sequence diagrams developed in Chapter 3 by defining how objects and components collaborate at runtime.

4.3 Design Goals

The design of the Offline Amharic First Aid Advisor is guided by well-established software engineering principles to ensure quality, reliability and usability. The main design goals are described below.

4.3.1 Modularity

The system is divided into independent modules such as NLP processing, content management, offline maps and user interface. Each module performs a specific function and can be developed, tested and maintained independently.

4.3.2 High Cohesion and Low Coupling

Each module focuses on a single responsibility (high cohesion) and interacts with other modules through well defined interfaces (low coupling). This improves system stability and simplifies future enhancements.

4.3.3 Usability and Accessibility

The system is designed with a simple, intuitive interface suitable for users with varying levels of digital literacy. Amharic language support, audio and text guidance and visual aids enhance accessibility for a wider population.

4.3.4 Performance and Reliability

Offline processing ensures fast response times and uninterrupted functionality during emergencies. Lightweight NLP models and optimized local databases improve performance even on low-end mobile devices.

4.3.5 Abstraction and Encapsulation

Internal processing details (such as NLP classification logic and database queries) are hidden from the user interface layer. This abstraction improves security and simplifies system management.

4.3.6 Reusability

Reusable components such as the NLP engine, database schema and UI elements are designed so they can be adapted for other health related or emergency response applications in the future.

4.3.7 Scalability and Maintainability

The design allows the system to scale by adding more emergency scenarios, health facilities, multimedia content or language support without redesigning the entire system.

4.4 Proposed system Architecture

4.4.1 Architectural Overview

The proposed Offline Amharic First Aid Advisor Using NLP and Offline Maps is implemented using a **Three-Tier (Layered) Architecture**, a widely accepted software design paradigm that separates the system into distinct logical layers. This architecture is chosen to ensure clear separation of concerns, modularity, maintainability, and reliability, which are critical for a safety-oriented mobile health application.

The Three-Tier Architecture divides the system into the following layers:

1. Presentation Tier (User Interface Layer)
2. Application Tier (Business Logic & Control Layer)
3. Data Tier (Offline Data & AI Layer)

This architectural approach aligns well with the system's offline-first requirement, as all tiers are deployed locally on the mobile device while still remaining logically independent. By isolating user interaction, processing logic, and data management, the system becomes easier to extend, debug, test, and maintain throughout its lifecycle.

4.4.2 Rationale for Architectural Choice

The Three-Tier Architecture was selected for the following reasons:

- Separation of Concerns: Each tier focuses on a specific responsibility, reducing system complexity.
- Maintainability: Updates to first aid content, NLP models, or UI design can be performed with minimal impact on other tiers.
- Scalability: New emergency scenarios, languages, or features can be added without restructuring the entire system.
- Reliability: Local execution minimizes dependency on external services and ensures continuous availability.
- Security & Privacy: Direct interaction between the user interface and the database is avoided, reducing data exposure risks.

4.4.3 Presentation Tier (User Interface Layer)

The Presentation Tier will represent the user-facing layer of the system and is responsible for all interactions between the user and the application. This tier ensures that emergency guidance is delivered in a clear, intuitive, and accessible manner, especially under stressful conditions.

Responsibilities:

- Capture user input through category selection, text input, and voice input in Amharic
- Display step-by-step first aid instructions
- Present images, icons, and audio playback for accessibility
- Show offline maps and nearby hospital listings
- Provide quick-access buttons for emergency calls

Key Features:

- Simple, emergency-focused UI design optimized for fast decision-making
- Amharic language support with large fonts and visual indicators
- Audio-guided instructions for low-literacy users
- Responsive layout suitable for various mobile screen sizes

Technologies:

- Flutter (Dart) for cross-platform mobile development
- Native device UI components for performance and reliability

Example Usage:

A user selects “Choking” from the emergency categories and immediately views illustrated, step-by-step instructions with optional audio guidance.

4.4.4 Application Tier (Business Logic Layer)

The Application Tier serves as the core processing layer of the system. It acts as an intermediary between the Presentation Tier and the Data Tier, handling all business rules, decision-making processes, and coordination of system components.

Responsibilities:

- Process user actions and navigation requests
- Execute Amharic NLP intent classification
- Match detected emergencies with appropriate first aid content
- Manage multimedia playback logic
- Calculate nearest hospitals using GPS data
- Coordinate offline map interactions and routing

Key Components:

- Emergency Classification Controller
- NLP Intent Processing Module
- Instruction Retrieval Manager
- Map & Location Controller
- Emergency Call Handler

Technologies:

- Dart-based application logic (Flutter)
- TensorFlow Lite integration for on-device NLP inference

Example Usage:

When a user says “ልጄ ለነቀው ለንጹሕ ልርዳው?” The application tier processes the NLP output, identifies the emergency as child choking, and retrieves the corresponding first aid guide.

4.4.5 Data Tier (Offline Data & AI Layer)

The Data Tier is the foundational layer responsible for storing, managing, and providing access to all persistent system data. This tier ensures that all critical information is available offline and remains consistent and secure.

Responsibilities:

- Store first aid instructions and emergency categories
- Manage multimedia resources (images, audio files)
- Store hospital and clinic location data
- Store and execute the offline Amharic NLP model
- Provide fast and reliable data retrieval

Key Features:

- Fully offline data availability
- Structured storage for efficient querying
- Optimized data access for low-end devices

Technologies:

- SQLite for structured local data storage
- TensorFlow Lite for NLP model execution
- Local file system for map tiles and media assets

Example Usage:

The system retrieves stored hospital location data from SQLite to display nearby clinics on the offline map.

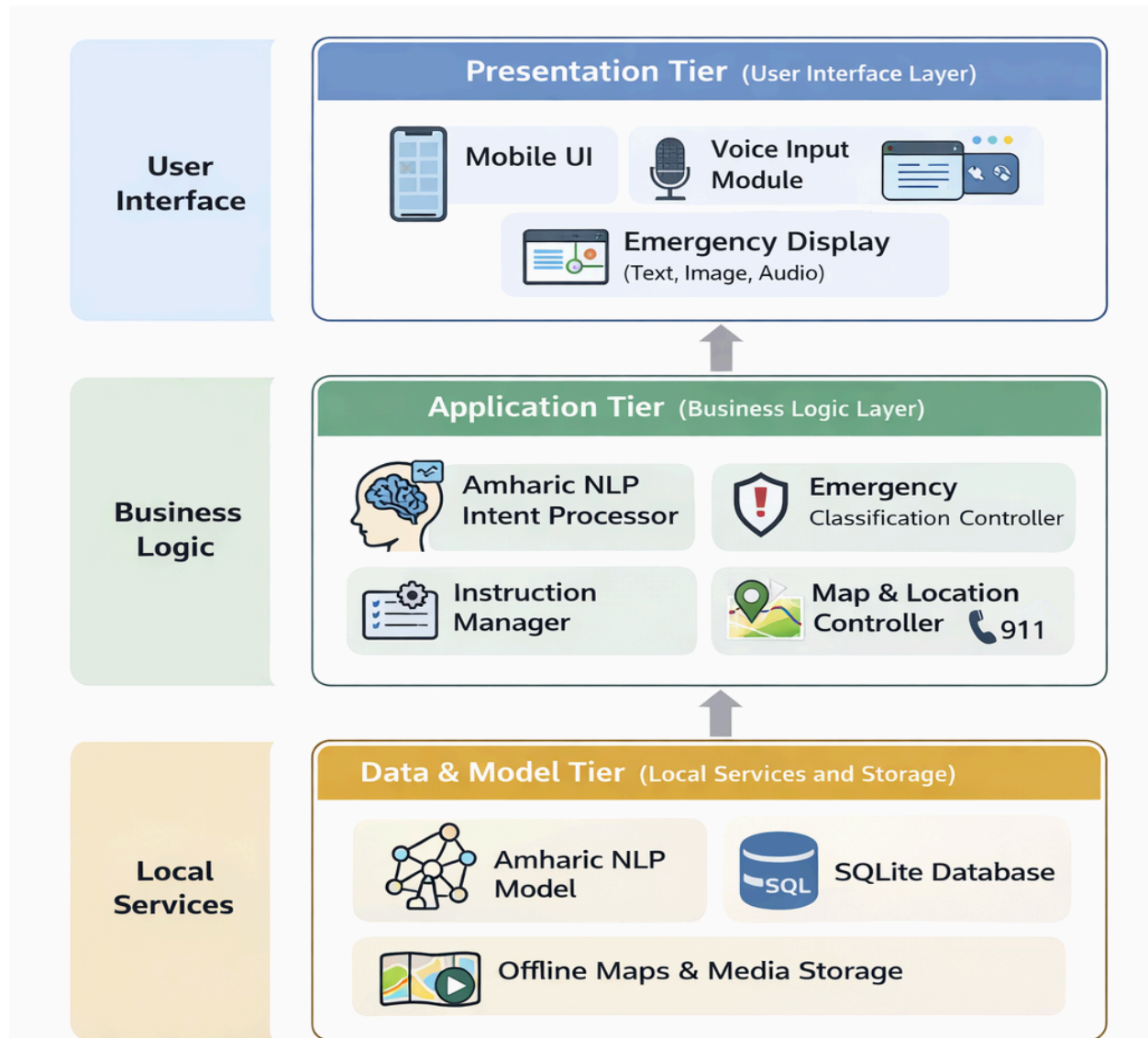


Figure 14 System Architecture

4.5 Subsystem Decomposition

4.5.1 Overview of Subsystem Decomposition

Subsystem decomposition breaks the system into smaller, cohesive units, each responsible for a specific functional domain. This approach enhances modularity, simplifies development, and allows independent testing of system components.

The proposed system is decomposed into the following subsystems:

1. User Interface Subsystem
2. First Aid Knowledge Management Subsystem
3. Amharic NLP Subsystem
4. Offline Map & Hospital Locator Subsystem
5. Emergency Communication Subsystem
6. System Utilities & Support Subsystem

Each subsystem exposes well-defined interfaces and communicates with other subsystems through the application logic layer.

4.5.2 Subsystem Hierarchy

Offline Amharic First Aid Advisor System

- UI Subsystem
- NLP Subsystem
- Knowledge Base Subsystem
- Map & Location Subsystem
- Emergency Communication Subsystem
- Utilities & Support Subsystem

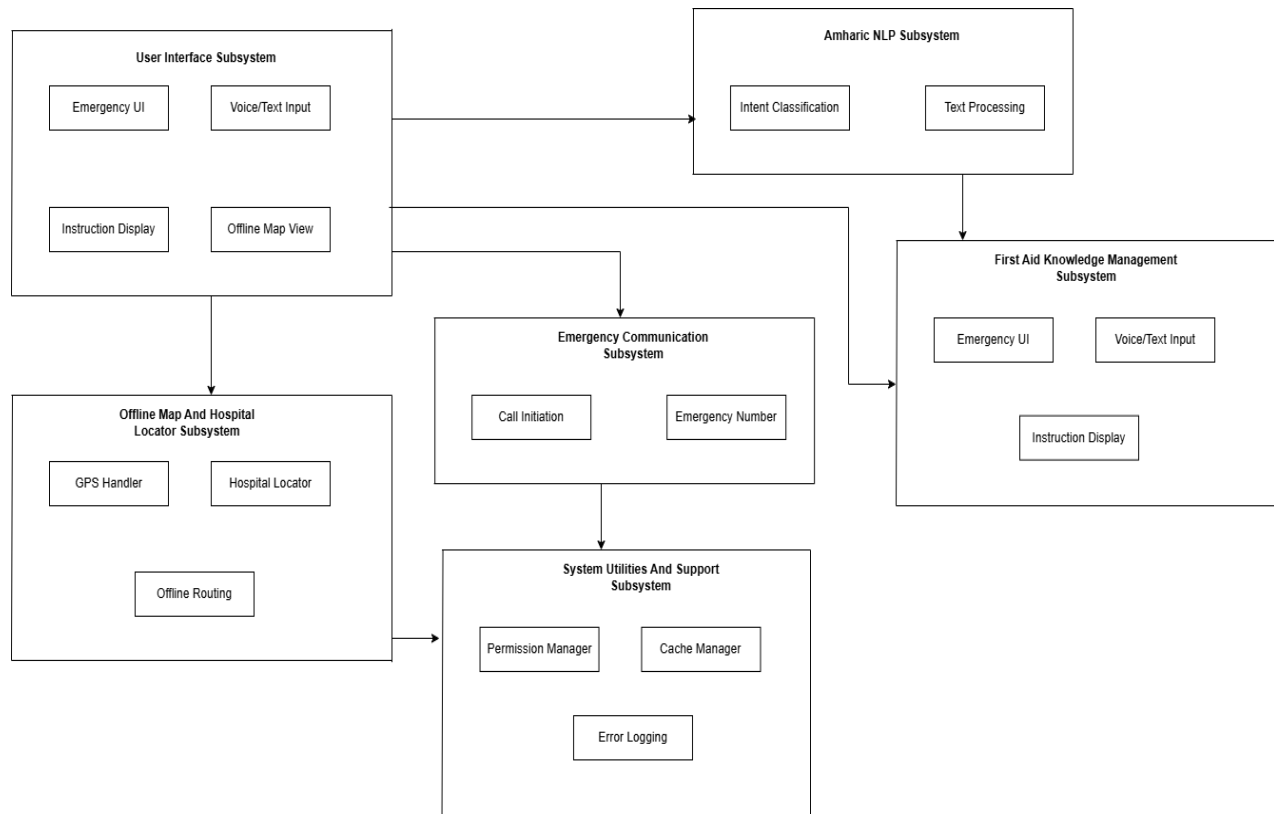


Figure 15 Subsystem Decomposition

4.6 Subsystem Description

4.6.1 User Interface (UI) Subsystem

Purpose:

Provides all visual interaction points between the user and the system.

Main Functions:

- Display emergency categories
- Accept text and voice input in Amharic
- Present step-by-step first aid guidance
- Display offline maps and hospital lists
- Provide quick-access emergency buttons

Interfaces:

- Communicates with Application Logic Layer APIs
- Receives processed data (instructions, locations)

Dependencies:

- Depends on Application Logic Subsystem

4.6.2 First Aid Knowledge Management Subsystem

Purpose:

Manages all first aid content and instructional data stored locally.

Main Functions:

- Store emergency categories and procedures
- Retrieve step-by-step instructions
- Manage multimedia resources (images, audio)
- Support bookmarking and content lookup

Data Managed:

- Emergency types
- Instruction steps
- Audio and image references

Dependencies:

- Depends on SQLite database
- Accessed by NLP and UI subsystems

4.6.3 Amharic NLP Subsystem

Purpose:

Interprets user queries expressed in natural Amharic language and identifies the emergency intent.

Main Functions:

- Process Amharic text input
- Process offline speech-to-text output
- Classify emergency intent
- Handle ambiguous queries by suggesting alternatives

Technologies:

- TensorFlow Lite NLP model

Dependencies:

- Depends on local NLP model files
- Interacts with Knowledge Management Subsystem

4.6.4 Offline Map & Hospital Locator Subsystem

Purpose:

Helps users locate nearby hospitals, clinics, and health centers without internet access.

Main Functions:

- Display offline map tiles
- Retrieve GPS location
- Load hospital locations from database
- Calculate nearest facilities
- Display basic route lines

Dependencies:

- Depends on GPS hardware
- Depends on local map tiles and hospital database

4.6.5 Emergency Communication Subsystem

Purpose:

Enables rapid communication with emergency services and health facilities.

Main Functions:

- Display emergency phone numbers (e.g., 907, 939)
- Initiate calls through device dialer
- Provide quick-access emergency call buttons

Dependencies:

- Depends on device phone services

4.6.6 System Support & Utilities Subsystem

Purpose:

Provides supporting services required for smooth system operation.

Main Functions:

- Permission handling (GPS, microphone, storage)
- Local caching and file management
- Error handling and fallback mechanisms
- Performance optimization

Dependencies:

- Interacts with all subsystems as a shared utility layer

4.7. Persistent Data Management

4.7.1. Overview

The Offline Amharic First Aid Mobile Application implements a comprehensive persistent data management system designed for complete offline functionality. All data is stored locally on the user's Android device using SQLite, a lightweight, embedded relational database that requires zero network connectivity. The system manages structured first aid knowledge, user preferences, bookmarks, and offline map data while ensuring data integrity, security, and efficient retrieval even in emergency situations with no internet access.

4.7.2. Data Storage Architecture

Primary Storage: SQLite Database

- Database Engine: SQLite 3.35+ (Android built-in)
- Storage Location: /data/data/com.offlinefirstaid/databases/first_aid.db
- File Size: Approximately 50-100 MB (compressed)
- Backup Strategy: Automatic backup to Android's Auto Backup for Apps
- Encryption: SQLite Encryption Extension (SEE) for sensitive data

Secondary Storage: File System

- Asset Files: Images, audio files, NLP models stored in assets/ folder
- MapTiles: Compressed PNG files in
storage/emulated/0/Android/data/com.offlinefirstaid/maps/
- Cache: Temporary NLP processing cache in app-specific cache directory

4.7.3. Data Models and Object-Relational Mapping

Object to Database Mapping

Category Object Mapping

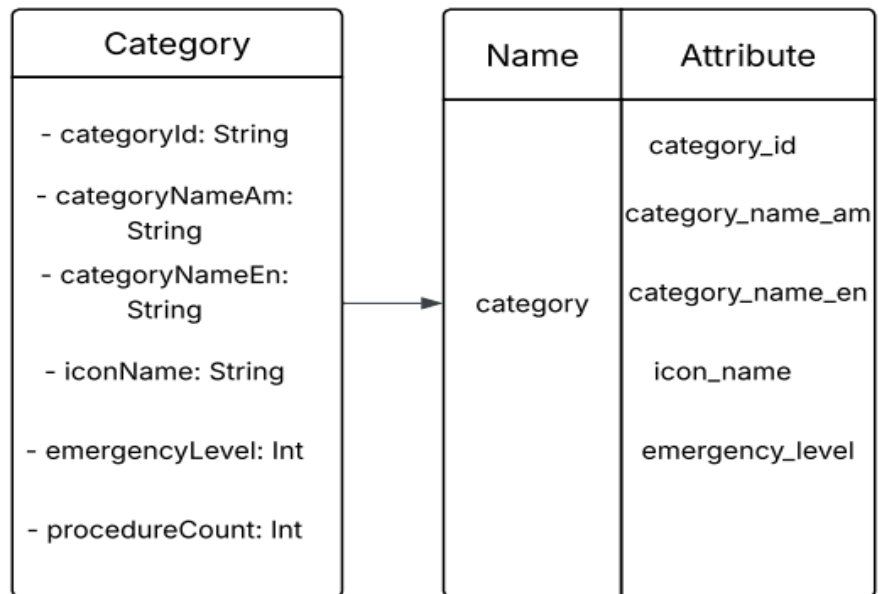


Figure 16 Category Object Mapping

Procedure Object

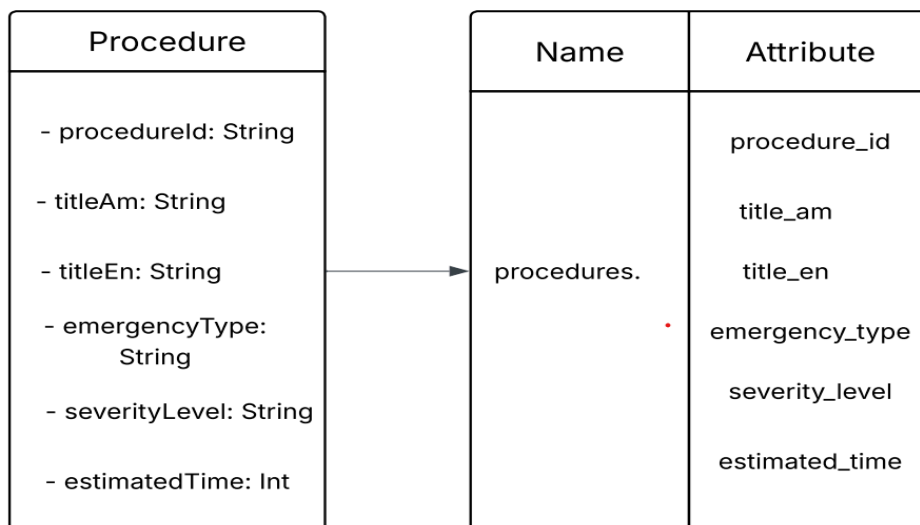


Figure 17 Procedure Object

Medical Facility Object Mapping

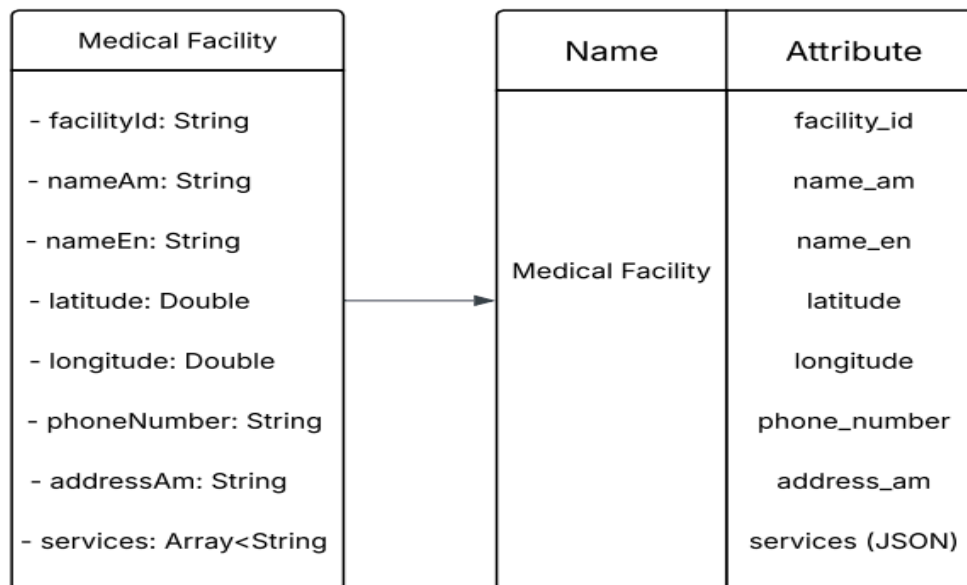


Figure 18 Medical Facility Object Mapping

4.7.4. Performance Considerations

1. Query Optimization Indexing Strategy:
 - Primary indexes on all ID columns
 - Composite indexes on (category_id, severity_level)
 - Full-text search index on Amharic text columns
 - Query Caching: Frequently accessed data cached in memory
 - Lazy Loading: Procedure steps loaded on demand
2. Memory Management
 - Pagination: Large results returned in pages (20 items per page)
 - Blob Storage: Images and audio stored as file references, not in database
 - Connection Pooling: Single database connection shared across app
3. Storage Efficiency
 - Data Compression: Text compression for Amharic content

- Binary Storage: Efficient storage of map tiles and NLP models
- Cleanup Routines: Automatic removal of temporary files

4.8. Component Diagram

Overview

In this Diagram components of the system will be wired showing that there is relation among components, management of the system, database and operations performed on databases such as security issues. This in some extent shows which components or objects will be accessed by whom and what type of security infrastructures it is using. The diagram is simulated below.

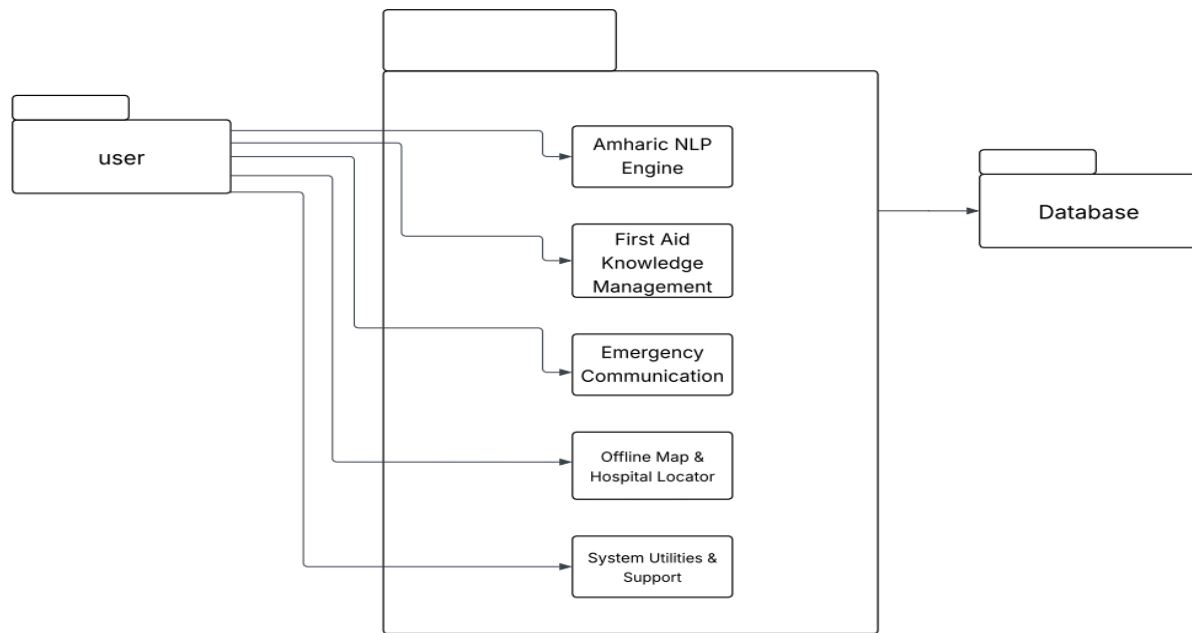


Figure 19 Component Diagram

4.9. Database Diagram

4.9.1 Overview

The database diagram represents the structure of the local data storage used by the Offline Amharic First Aid Advisor. Since the system is designed to operate fully offline, all required data

is stored locally on the user's mobile device using an SQLite database. The database supports fast access to first aid instructions, emergency categories, hospital information, multimedia references, and bookmarked guides.

The database design focuses on:

- Simplicity and efficiency
- Offline reliability
- Minimal storage usage
- No personal user data storage

4.9.2 Database Technology

- **Database Type:** SQLite
- **Storage Location:** Local mobile device storage
- **Connectivity:** Fully offline
- **User Accounts:** Not required
- **Data Persistence:** Local and permanent unless the application is uninstalled

SQLite is selected because it is lightweight, fast, reliable, and well-suited for offline Android applications.

4.9.3 Database Entities and Description

1. Category Table

Field Name	Data Type	Description
category_id	INTEGER (PK)	Unique identifier for each first aid category
name	TEXT	Name of the emergency category (e.g., Burns, Bleeding)
description	TEXT	Brief description of the category

Table 11 Category Table

2. Guide Table

Field Name	Data Type	Description
guide_id	INTEGER (PK)	Unique identifier for each first aid guide
category_id	INTEGER (FK)	References the related category
title	TEXT	Title of the first aid guide
steps	TEXT	Step-by-step first aid instructions
image_path	TEXT	Local file path of instructional images
audio_path	TEXT	Local file path of audio guidance

Table 12 Guide Table

3. Hospital Table

Stores information about hospitals and clinics.

Field Name	Data Type	Description
hospital_id	INTEGER (PK)	Unique identifier for hospital or clinic
name	TEXT	Name of the hospital or clinic
phone	TEXT	Contact phone number
latitude	REAL	Geographic latitude
longitude	REAL	Geographic longitude

type	TEXT	Facility type (Hospital / Clinic / Health Post)
------	------	---

Table 13 Hospital Table

4. Bookmark Table

Stores bookmarked first aid guides.

Field Name	Data Type	Description
bookmark_id	INTEGER (PK)	Unique identifier for bookmark
guide_id	INTEGER (FK)	References bookmarked guide
created_at	TEXT	Date and time bookmark was created

Table 14 Bookmark Table

Entity Relationship Correspondence

Entity A (Parent)	Entity B (Child)	Relationship Type	Explanation
Category	Guide	One-to-Many (1:N)	One first aid category contains multiple guides
Guide	Bookmark	One-to-Many (1:N)	One guide can be bookmarked multiple times
Guide	Category	Many-to-One (N:1)	Each guide belongs to only one category

Table 15 Entity Relationship Correspondence

One **Category** can have many **Guides** (1-to-Many)

- One **Guide** can be bookmarked multiple times (1-to-Many)
- Bookmarks reference **Guides**, not users
- Hospitals are independent entities and not linked to users

4.9.5 Database Diagram Description

The database diagram visually illustrates:

- Tables as entities
- Primary keys (PK)
- Foreign key (FK) relationships
- Clear one-to-many relationships

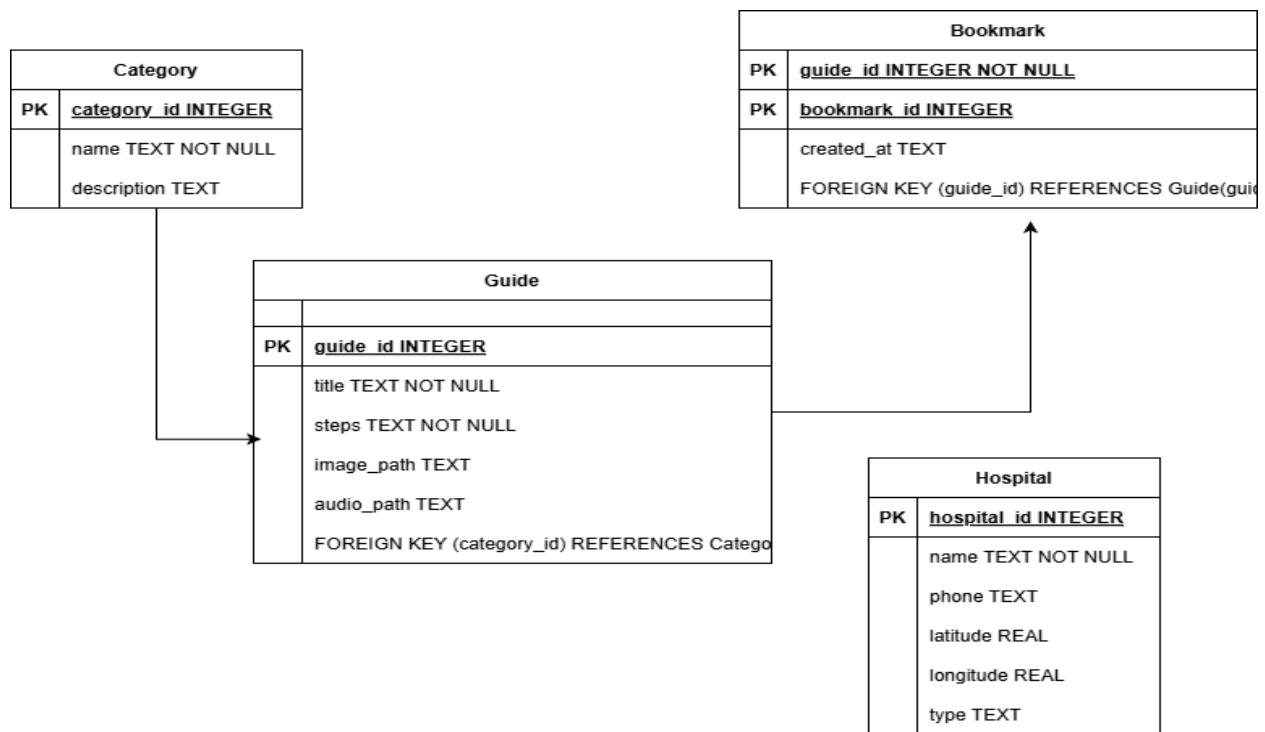


Figure 20 Database Diagram

4.10 Access Control

4.10.1 Overview

Access control defines how the system regulates access to its features, data, and device resources. The offline Amharic First Aid Advisor follows a permission-based access control model rather than a traditional authentication model.

Since the application does not require user accounts, login, or registration, access control is limited to device-level permission only.

4.10.2 Access Control Model

The system uses a Simple Permission-Based Access Control (PBAC) approach:

- No authentication (no usernames or passwords)
- No authorization roles
- No user identity management
- Controlled access through Android system permissions

This approach aligns with emergency usage scenarios where speed and simplicity are critical.

4.10.3 Device-Level Permissions

1. Microphone Permission

- Purpose: Accept offline Amharic speech input
- Used by: NLP Module
- Activated when: User taps voice input
- Data Handling: Audio processed locally and not stored

2. Location (GPS) Permission

- Purpose: Retrieve the use's current location for offline hospital mapping
- Used by: Offline Map Module
- Activated when: User opens "Nearby Hospitals"
- Data Handling: Location data is used temporarily and not stored

3. Phone Call Permission

- Purpose: Call emergency services (907/939) or clinics
- Used by: Emergency Service Module
- Activated when: User presses “Call Now”
- Data Handling: No call data is stored

4.10.4 Data Privacy and Security

- No personal data collection
- No external data transmission
- No cloud storage
- All processing performed locally
- Database accessible only to the application

This ensures maximum privacy, especially in emergency situations.

4.10.5 Security Considerations

- SQLite database is sandboxed by the Android OS
- No exposure to network-based attacks
- Minimal attack surface due to offline operation
- System stability maintained during emergency usage

4.10.6 Alignment with System Requirements

The access control approach supports the system's non-functional requirements by ensuring offline operation, protecting user privacy, and minimizing interaction complexity. By avoiding authentication and cloud services, the system guarantees immediate access to emergency guidance while maintaining data security and reliability.

4.10.7 Summary

The access control strategy prioritizes speed, privacy and reliability. By eliminating user authentication and relying solely on device permissions, the system ensures that users can access critical first aid guidance immediately during emergencies without unnecessary barriers.

Chapter 5: Implementation

5.1 Overview

This chapter describes the planned implementation phase of the Offline Amharic First Aid Advisor Using NLP and Offline Maps. The implementation phase will focus on translating the system design and requirements into a fully functional mobile application that operates reliably without internet connectivity.

The system will be implemented as an **offline-first mobile application** using **Flutter** as the primary development framework. All core functionalities - including first aid guidance, speech-based emergency assistance, offline maps, and emergency contact features - will be executed locally on the user's device to ensure continuous availability during emergency situations.

The implementation will be organized into three main technical components:

- **Client-side implementation**, responsible for user interaction, navigation, and presentation of first aid instructions
- **Local data management**, responsible for storing and retrieving first aid content, hospital data, and multimedia resources using a local database
- **Machine learning integration**, responsible for offline Amharic speech processing and emergency intent classification using an on-device NLP model

No server-side or cloud-based backend will be implemented, as the system is intentionally designed to function entirely offline. This design choice ensures reliability in low-connectivity environments and aligns with the project's goal of supporting emergency use cases where network access cannot be guaranteed.

During implementation, a functional prototype will be developed iteratively. Each iteration will focus on validating design assumptions, testing offline performance, and refining usability based on simulated emergency scenarios. The final implementation will integrate all system components into a cohesive, maintainable, and scalable mobile application.

5.2 Coding Standard

To ensure code quality, maintainability, and long-term extensibility, the implementation of the system will follow established coding standards and best practices appropriate for Flutter and Dart-based development.

Naming Conventions

- Classes will be named using **PascalCase** (e.g., EmergencyController, FirstAidGuide)
- Variables and methods will use **camelCase** (e.g., loadGuides(), currentLocation)
- Constants will be written in **UPPER_CASE** (e.g., MAX_AUDIO_DURATION)
- File names will follow **snake_case** (e.g., emergency_handler.dart)

Code Structure

- The project will be organized using a **layered architecture**, separating:
 - UI components
 - Business logic
 - Data access and services
- Flutter widgets will be kept lightweight, with business logic handled in dedicated controller or service classes
- Reusable components (widgets, helpers, utilities) will be modularized to reduce duplication

Documentation and Comments

- All major classes and methods will include inline documentation comments explaining their purpose and functionality

- Complex logic, especially related to NLP processing and emergency classification, will be clearly commented for readability
- Public interfaces and critical modules will be documented to support future maintenance or extension

Best Practices

- Flutter's recommended performance practices will be followed to ensure smooth execution on low-end devices
- Asynchronous operations (e.g., database access, audio playback) will be handled safely using Dart's `async/await` pattern
- Error handling will be implemented to gracefully manage unexpected inputs or system limitations
- Code formatting will follow Dart's official style guidelines to maintain consistency across the project

Adhering to these coding standards will help ensure that the system remains readable, reliable, and easy to extend, while also supporting collaborative development and academic evaluation.

5.3 Development Tools

This section of the document comprehensively explains development tools that will be utilized throughout the implementation phase of the Offline Amharic First Aid Advisor project. These tools including integrated development environments (IDEs), version control systems, debugging tools, design software and testing utilities

5.3.1 Integrated Development Environments (IDEs)

The team will employ two primary IDEs that complemented each other throughout the development lifecycle:

- **Android Studio:** Served as the primary IDE for Flutter development due to its native Android emulator, comprehensive Flutter plugin support, and integrated debugging tools.

- Visual Studio Code (VS Code): will be used as a secondary IDE, particularly valued for its lightweight nature, extensive extension marketplace, and excellent support for Python development. VS Code was instrumental during the NLP model development phase, allowing seamless switching between Dart (for mobile app) and Python (for AI/ML) workflows within the same development environment.

5.3.2 Version Control and Collaboration Systems

Git with GitHub was used for version control and team collaboration, enabling efficient code management, issue tracking, and collaborative development through pull requests.

5.3.3 Design and Prototyping Tools

- Figma will be used as the primary UI/UX design and prototyping tool, enabling real-time collaboration, interactive mockups, and early user testing to create an intuitive interface suited to users with varying levels of technical literacy.
- Flutter DevTools: This suite of debugging and performance tools will be indispensable for identifying UI rendering issues, memory leaks, and performance bottlenecks. The timeline view helped optimize widget rebuilds, while the memory profiler ensured the application remained efficient on devices with limited RAM, a critical consideration for the target user base.
- Android Emulator & Physical Devices: A combination of virtual and physical testing environments will be maintained. Multiple Android emulator configurations simulated various device specifications and OS versions, while physical testing on actual mid-range and low-end Android devices validated real-world performance and usability.
- Postman will be used to test data update endpoints and validate compressed update packages during development.

5.3.5 AI/ML Development and Model Optimization Tools

TensorFlow Lite & ONNX Runtime: These lightweight inference engines were evaluated for deploying the trained NLP models on mobile devices. TensorFlow Lite was ultimately selected for its better integration with the Flutter ecosystem through the tflite-flutter package, though ONNX Runtime remained a viable backup option for performance comparison.

Hugging Face Transformers: This library accelerates the NLP development by providing pre-trained multilingual models that could be fine tuned on the Amharic first aid dataset, significantly reducing development time compared to building models from scratch.

5.3.6 Documentation and Diagramming Tools

Google Docs & Microsoft Word: Facilitated collaborative documentation of requirements, test cases and user manuals. The version history and commenting features enabled efficient peer review of all project documentation.

Draw.io/Lucid: Used for creating system architecture diagrams, data flow charts, and database schemas. Its integration with Google Drive allowed real-time collaborative diagram editing, ensuring all team members had access to the most current system design representations.

5.3.7 Project Management and Communication Tools

Slack: its being used as the communication channel for daily stand ups, quick questions, and file sharing, reducing email overload and keeping all project discussions organized in dedicated channels.

5.4 Prototype

This section describes the development of the prototype for the Offline Amharic First Aid Advisor. The prototype was created to visualize core system functionality, validate design decisions, and evaluate usability before full implementation. It focuses on demonstrating essential user interactions in emergency and non-emergency scenarios while reflecting offline usage constraints.

The prototype was guided by the following objectives:

- Enable fast access to first-aid guidance during emergencies
- Minimize user interaction steps under stress
- Support offline-first usage scenarios
- Present content clearly in the Amharic language

- Ensure usability on low-end mobile devices

Prototype Screens

1. Welcome Screen

The Welcome Screen introduces the application and briefly explains its purpose. It includes a medical disclaimer stating that the application provides first-aid guidance and does not replace professional medical care. Users can proceed directly to the Home Screen without registration.

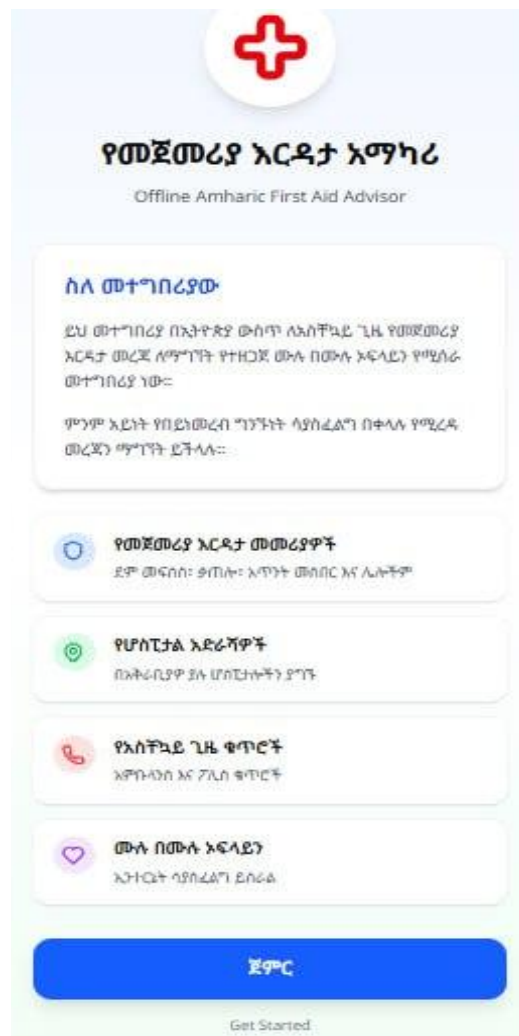


Figure 21 Welcome Screen

2. Home Screen

The Home Screen serves as the main navigation hub of the application. It provides quick access to emergency assistance, first-aid categories, offline hospital location, and emergency call shortcuts. The layout prioritizes critical actions to support rapid decision-making.

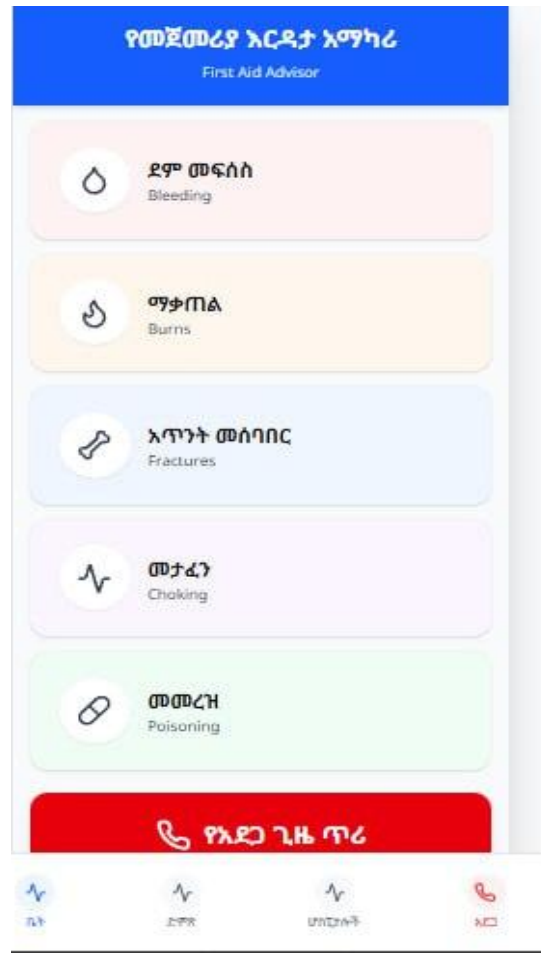


Figure 22 Home Screen

3. Emergency Assistance Screen

This screen allows users to enter a short text description of an emergency in Amharic. The system processes the input using lightweight intent classification and directs the user to the appropriate first-aid guidance.



Figure 23 Emergency Assistance Screen

4. Offline Hospital Locator Screen

The Offline Hospital Locator Screen displays nearby medical facilities using an offline map or list-based view. Users can identify the nearest hospitals or clinics without internet connectivity.

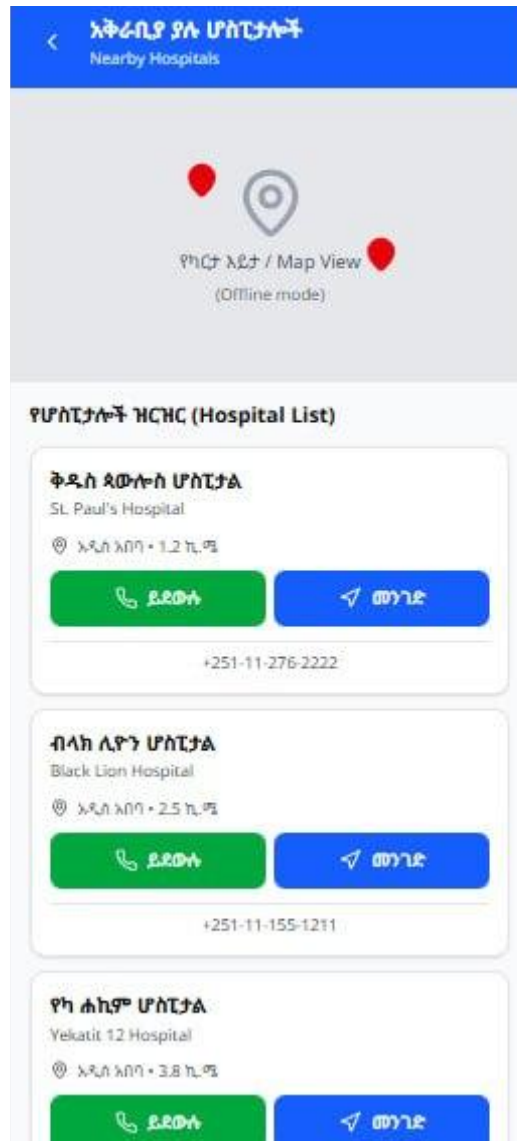


Figure 24 Offline Hospital Locator Screen

Prototype Validation

The prototype was evaluated through iterative reviews focusing on usability, navigation clarity, and alignment with functional requirements. Design improvements included simplifying screen layouts, reducing text density, and ensuring consistent navigation across all screens. The final prototype serves as a reference for system implementation and testing.

5.5 Implementation Detail

The system will be implemented entirely on the client side using Flutter, with all core functionalities running locally on the user's device. No backend server or remote database will be used, ensuring uninterrupted operation during emergency situations.

5.5.1 Client-Side Implementation

The client-side will be implemented as a **cross-platform mobile application using Flutter**. It is responsible for presenting the user interface, handling user interactions, capturing voice input, and displaying first aid guidance.

Technologies to be Used

- **Framework:** Flutter
- **Programming Language:** Dart
- **UI Design:** Flutter widgets and Material Design
- **Local Storage:** SQLite
- **Audio Services:** Device microphone and audio player APIs
- **Offline Maps:** Preloaded map data and clinic location datasets

Client-Side Architecture

The Flutter application will follow a **widget-based and layered architecture**, separating UI components from business logic.

Main screens include:

- Home screen with first aid categories
- Emergency voice input interface (Amharic)
- First aid instruction viewer
- Offline clinic locator
- Emergency contact and calling interface

Client-Side Functionality

- Offline browsing of first aid instructions
- Voice-based emergency queries
- Audio playback for hands-free guidance
- Offline navigation to clinics
- Direct calling of emergency contacts

All processing will be handled locally to ensure fast response time and reliability.

5.5.2 Local Data Management (No Server-Side Implementation)

The system will not include a server-side implementation. All required data will be stored and managed locally on the device using a SQLite database.

Stored Data Includes:

- First aid categories and procedures
- Educational first aid content
- Clinic and hospital information
- Emergency contact numbers
- NLP keyword mappings

This approach ensures full functionality without internet connectivity and improves data privacy and system reliability.

5.5.3 Machine Learning Model

Machine learning components enhance system intelligence while remaining lightweight and fully offline.

Offline Speech-to-Text (STT)

- Uses a pre-trained offline STT model
- Supports Amharic language
- Runs entirely on-device

Intent Detection

- Keyword-based and rule-based classification
- Optimized for emergency scenarios

Evaluation Criteria

- Recognition accuracy
- Response time
- Offline reliability

The implementation prioritizes offline reliability, simplicity, and fast emergency response. By using Flutter, local SQLite storage, and lightweight offline machine learning techniques.

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